LATENT COVERAGE FROM NOISY CALIBRATION

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Supplementary Material

Numbers of sections, theorems, equations, tables and figures without the prefix S refer to the main paper. Section S1 collects facts used throughout; Sections S2 and S3 prove the results of Sections 3 and 4 of the main paper; Section S4 treats a known Gaussian noise variance, which the main paper mentions only in its discussion; Section S5 describes the certificates in ball arithmetic; Sections S6 and S7 give the design and the full results of the numerical studies and of the application; Section S8 maps every result to the code that produces it. Each numerical statement says how it is supported: proved, certified in ball arithmetic, computed in double precision, or estimated by simulation. Table S1 shows where each result of the main paper is proved.

S1. Preliminaries

Notation. The noise of a unit is $e = \sigma\epsilon$, with $\sigma > 0$ and ϵ a standardized noise variable whose law lies in $\mathcal{C}$; $\mathcal{N} = \{\sigma Z : \sigma > 0\}$ is the class of Gaussian noise laws, written $\mathcal{C}$ in Theorem 2, so that $\Psi_{\mathcal{N}}$, $\lambda_{\mathcal{N}}$ and $p_{\mathcal{N}}$ are

Table S1: Where the results of the main paper are supported

main-paper result	Supplement	support
Proposition 2	main text	proved
two-sided level $p_{\mathcal{C}}(q)$	§ S2.1, Lemmas S2–S3	proved
$\Psi_{\mathcal{C}}(q)$ cannot be lowered	§ S2.2	proved
Theorem 1(i)	§ S2.3, Proposition S1	proved; values in ball arithmetic
Theorem 1(ii)	§ S2.4, § S5	proved; bounds certified in ball arithmetic
Theorem 1(iii)	§ S2.5	proved
Theorem 2	§ S2.6	proved; values in ball arithmetic
Proposition 1	§ S2.7	proved; examples in closed form
Theorem 3; ranks of Table 2, including its first row	§ S3.1	proved; ranks computed exactly
Proposition 3	main text; § S3.1	proved
Corollary 1 and the remarks after it	§ S3.2, Lemmas S5–S6	proved; exact check in § S6.5
Proposition 5	§ S3.3	proved; exact check in § S6.6
Proposition 4, Table 4	§ S3.4, § S6.7	(i), (ii), (iii) proved; values for $\zeta < 1$ numerical
known variance: Theorem S1, the 1.16% bound	§§ S4.2–S4.4	proved; c_q certified in ball arithmetic
known variance: the location of the mean	§ S4.6, Theorems S2–S3	proved; constants in double precision
known variance: the noisy level that removes the widening	§ S4.5, Proposition S4	proved
rules with known variances (§ 5.3)	§ S4.7	proved; radius computed in double precision
the shift of the new residual	§ S3.5, Proposition S2	proved
Tables 2–6, Figure 1	§ S6	exact computation; simulation
the application (§ 6)	§ S7	simulation on a finite population

the Gaussian levels; X_u denotes an exponential variable with rate $u > 0$, and X_1 a standard exponential variable; Z is standard normal, Φ and ϕ its distribution function and density; e is the base of the natural logarithm and Q_q the lower q-quantile. A law is identified with its distribution function

where convenient. Table S2 lists the symbols used across sections; letters such as a, b, c, m, r and s also serve as local variables, defined where they are used.

Table S2: Symbols used across sections

symbol	meaning
W, $V = W + e$	latent residual and noisy score; $e = \sigma\epsilon$
$\mathcal{B}$, $\mathcal{L}$	bi-log-concave and log-concave laws
$\mathcal{C}$, $\mathcal{N}$	a class of standardized noise laws; the Gaussian class
$\Psi_{\mathcal{C}}(q)$	one-end level (3.1)
$\lambda_{\mathcal{C}}$, $\vartheta_{\mathcal{C}}$	inverse level and largest one-end latent failure (§ S2.1)
$p_{\mathcal{C}}(q)$	two-sided sufficient level (3.2)
$\beta_{\mathcal{C}}$	$\inf_{\epsilon\in\mathcal{C}} \mathrm{pr}(\epsilon \geq 0)$
$\Gamma(q, s)$	$E\{\min(1, q\mathrm{e}^{-sZ})\}$, Gaussian noise of scale s (3.3)
$m(b)$, $A(b)$	$\phi(b)/\Phi(b)$ and $1 - \mathrm{var}(Z \mid Z \leq b)$ (§ S2.6)
K, k, δ, q	calibration size, rank, $1-$ reliability, latent coverage
Q_q, $T = \lvert V\rvert_{(k)}$	lower q-quantile; noisy threshold
$x = D/t^2$, $R_{p,q}(x)$	noise-to-threshold ratio; sharp radius (S4.1)
$m_u(b)$, $b_u(p)$	noisy one-end coverage of $b - X_u$ with unit Gaussian noise, and its inverse (§ S4.2)
$c_{p,q}$, $C_{p,q}$	one-sided and two-sided constants (Lemmas S7 and S9)

Bi-log-concave laws. $\mathcal{B}$ is the class of laws that are point masses or have a continuous distribution function F with $\log F$ and $\log(1 - F)$ concave (Dümbgen, Kolesnyk, and Wilke, 2017); $\mathcal{L} \subset \mathcal{B}$. Both classes are closed under translation and under multiplication by a nonzero constant: a negative constant exchanges the roles of $\log F$ and $\log(1-F)$. The following facts are used throughout.

1. A law in $\mathcal{B}$ that is not a point mass has no atom and has a density f that is positive on the interior of its support, the hazard $f/(1 - F)$ is nondecreasing and f/F is nonincreasing, and $f' \leq f^2/F$ where f is

differentiable (Dümbgen, Kolesnyk, and Wilke, 2017).

2. $\mathcal{B}$ is closed under convolution with a log-concave law. If $\log F_X$ is concave and g is a log-concave density, $(t, y) \mapsto F_X(t - y)g(y)$ is log-concave on $\mathbb{R}^2$, so $F_{X+Y}(t) = \int F_X(t-y)g(y)\,dy$ is log-concave by the theorem of Prékopa (1973), and likewise $1 - F_{X+Y}$.

3. $\mathcal{B}$ is closed under weak convergence (Saumard, 2019): concavity of $\log F_n$ and $\log(1 - F_n)$ passes to the limit at continuity points, and a limit that is not a point mass has no atom, because a concave function is continuous on the interior of its domain.

4. If X has a continuous distribution function F with $\log F$ concave and a finite mean, then $F\{E(X)\} \geq \mathrm{e}^{-1}$: $F(X)$ is uniform, so $E\{\log F(X)\} = -1$, and Jensen's inequality gives $\log F\{E(X)\} \geq E\{\log F(X)\}$. If also $\log(1 - F)$ is concave, $1 - F\{E(X)\} \geq \mathrm{e}^{-1}$ in the same way.

5. For the exponential tail $b - X_u$, the distribution function is $\min\{1, \mathrm{e}^{u(y-b)}\}$; it is log-concave, and if $\log F_Y$ is concave and u is a supergradient of $\log F_Y$ at a point v with $F_Y(v) = q$, then $F_Y(y) \leq \min\{1, q\mathrm{e}^{u(y-v)}\}$ for all y, that is, Y is stochastically larger than $v + u^{-1}\log(1/q) - X_u$.

A bimodal law in $\mathcal{B}$. The bimodal latent law of § 5.2 is the mixture $\pi N(m_1, s^2) + (1-\pi)N(m_2, s^2)$ with mean zero, variance one, $\pi = 0.4$ and $(m_2 - m_1)/s = 2.5$, that is $m_1 = -0.94868$, $m_2 = 0.63246$ and $s = 0.63246$. It is bimodal, since bimodality starts at a separation of about 2.443 for $\pi = 0.4$ (numerical), and it lies in $\mathcal{B}$ by the following lemma and the certificate after it; with $\pi = 0.4$ the mixture stays in $\mathcal{B}$ up to a separation of about 2.59 (numerical).

Lemma S1. *Let f be the density of $\pi N(m_1, s^2) + (1-\pi)N(m_2, s^2)$ with $m_1 < m_2$ and $r = (m_2 - m_1)/s > 2$, let $P(w)$ be the posterior probability of the first component, and let $a < b$ solve $P(a) = \frac{1}{2} + h$ and $P(b) = \frac{1}{2} - h$, with $h = (1/4 - 1/r^2)^{1/2}$. The mixture is in $\mathcal{B}$ if $f'F < f^2$ on $[a, m_2]$ and $-f'(1-F) < f^2$ on $[m_1, b]$.*

Proof. Since $(\log f)'' = -s^{-2} + P(1-P)r^2 s^{-2}$ and P decreases, f is log-concave on $(-\infty, a]$ and on $[b, \infty)$. On $(-\infty, a]$, $F(t) = \int 1(y \le t) f(y) 1(y \le a)\, dy$ is the marginal of a log-concave function of (t, y), so $\log F$ is concave there (Prékopa, 1973); likewise $\log(1-F)$ on $[b, \infty)$. Where $f' \le 0$, in particular on $[m_2, \infty)$, $f'F \le f^2$ is trivial, and where $f' \ge 0$, in particular on $(-\infty, m_1]$, so is $-f'(1-F) \le f^2$. With the two hypotheses, $(\log F)'' = (f'F - f^2)/F^2 \le 0$ and $\{\log(1-F)\}'' = \{-f'(1-F) - f^2\}/(1-F)^2 \le 0$ on the whole line. □

For the law above, the two hypotheses are proved in Arb ball arithmetic on 128 boxes, with the ends a and b rounded outward (§ S5); the largest values of $(f'F - f^2)/f^2$ on $[a, m_2]$ and of $\{-f'(1-F) - f^2\}/f^2$ on $[m_1, b]$ are -0.141 and -0.864.

S2. Coverage transfer

This section proves the results of § 3.

S2.1 Both ends of the interval

Let $\lambda_{\mathcal{C}}(p') = \sup\{q' \in (0,1) : \Psi_{\mathcal{C}}(q') < p'\}$, with $\sup \emptyset = 0$, and $\vartheta_{\mathcal{C}}(a) = 1 - \lambda_{\mathcal{C}}(1-a)$; both are nondecreasing. Let $\beta_{\mathcal{C}} = \inf_{\epsilon \in \mathcal{C}} \mathrm{pr}(\epsilon \geq 0)$. Since $\min(1, q'\mathrm{e}^{-\epsilon}) \leq q'$ when $\epsilon \geq 0$, $\Psi_{\mathcal{C}}(q') \leq 1 - (1-q')\beta_{\mathcal{C}}$. If $\beta_{\mathcal{C}} > 0$, as for Gaussian and symmetric unimodal noise, where $\beta_{\mathcal{C}} = 1/2$, and for mean-zero log-concave noise, where $\beta_{\mathcal{C}} = \mathrm{e}^{-1}$ by fact 4 of § S1, then $\Psi_{\mathcal{C}}(q') < 1$ for every $q' < 1$, so $\lambda_{\mathcal{C}}(1) = 1$ and $\vartheta_{\mathcal{C}}(0) = 0$. If instead $\Psi_{\mathcal{C}}(q') = 1$ for some $q' < 1$, Proposition 2 shows that no noisy level below 1 gives latent coverage q' at one end, so no transfer is possible; this is the case of mean-zero unimodal noise, Theorem 1(iii).

Lemma S2. *Let $\beta_{\mathcal{C}} > 0$, $W \in \mathcal{B}$, $t > 0$ and $e = \sigma\epsilon$ with $\sigma > 0$ and the law of ϵ in $\mathcal{C}$, and let $a_R = \mathrm{pr}(W + e > t)$ and $a_L = \mathrm{pr}(W + e < -t)$.*

Then $\mathrm{pr}(W > t) \le \vartheta_{\mathcal{C}}(a_R)$ *and* $\mathrm{pr}(W < -t) \le \vartheta_{\mathcal{C}}(a_L)$. *Consequently, if* $\mathrm{pr}(|W + e| \le t) \ge p > p_{\mathcal{C}}(q)$, *then* $\mathrm{pr}(|W| \le t) \ge q$.

Proof. The law of $Y = (W - t)/\sigma$ is in $\mathcal{B}$, so $\log F_Y$ is concave, and $\mathrm{pr}(Y + \epsilon \le 0) = 1 - a_R$. For every q' with $\Psi_{\mathcal{C}}(q') < 1 - a_R$, Proposition 2 gives $F_Y(0) \ge q'$, so $\mathrm{pr}(W \le t) \ge \lambda_{\mathcal{C}}(1 - a_R)$, which is the first bound. The law of $Y' = (-W - t)/\sigma$ is in $\mathcal{B}$ too, and $-\epsilon$ has a law in $\mathcal{C}$, which is closed under negation; as $Y' - \epsilon \le 0$ exactly when $W + e \ge -t$, $\mathrm{pr}(Y' - \epsilon \le 0) = 1 - a_L$, and the same argument gives $\mathrm{pr}(W \ge -t) = F_{Y'}(0) \ge \lambda_{\mathcal{C}}(1 - a_L)$, which is the second bound. Finally, the property in the definition (3.2) of $p_{\mathcal{C}}(q)$ is monotone in p, so it holds at every $p > p_{\mathcal{C}}(q)$; as $a_L + a_R = 1 - \mathrm{pr}(|W + e| \le t) \le 1 - p$, it gives $\mathrm{pr}(|W| > t) \le \vartheta_{\mathcal{C}}(a_L) + \vartheta_{\mathcal{C}}(a_R) \le 1 - q$. □

In the proof of Theorem 3 Lemma S2 is applied to one unit at a time, so the noise laws and scales may differ between units. Since $\vartheta_{\mathcal{C}}$ is nondecreasing, the condition in (3.2) can be checked on finitely many splits (a_L, a_R), as in § S5; a certificate replaces $\lambda_{\mathcal{C}}$ by a smaller certified level, which only enlarges $\vartheta_{\mathcal{C}}$ and so keeps the bound valid.

Lemma S3. *Suppose that* $\Psi_{\mathcal{C}} = 1/2$ *on* $(0, q_0]$ *and that* $\Psi_{\mathcal{C}}$ *is continuous, strictly increasing and convex on* $[q_0, 1)$, *with* $\Psi_{\mathcal{C}}(q) \to 1$ *as* $q \to 1$. *Then* $p_{\mathcal{C}}(q) \le \Psi_{\mathcal{C}}(q)$ *for every* $q \in (q_0, 1)$, *and noisy coverage* $p \in (1/2, 1)$ *at*

threshold t guarantees latent coverage $\lambda_{\mathcal{C}}(p) = \Psi_{\mathcal{C}}^{-1}(p)$ at t for every $W \in \mathcal{B}$.

Proof. On $(1/2, 1)$, $\lambda_{\mathcal{C}}$ is the inverse of the increasing convex function $\Psi_{\mathcal{C}}$ restricted to $(q_0, 1)$, so it is increasing and concave, and $\lambda_{\mathcal{C}}(1) = 1$. Hence $\vartheta_{\mathcal{C}}(a) = 1 - \lambda_{\mathcal{C}}(1-a)$ is convex on $[0, 1/2)$ with $\vartheta_{\mathcal{C}}(0) = 0$, and therefore superadditive there: for $a, b \geq 0$ with $a + b < 1/2$, convexity gives $\vartheta_{\mathcal{C}}(a) \leq \{a/(a+b)\}\vartheta_{\mathcal{C}}(a+b)$ and the same for b. With $a_L + a_R \leq 1 - p < 1/2$, Lemma S2 and monotonicity give $\mathrm{pr}(|W| > t) \leq \vartheta_{\mathcal{C}}(a_L) + \vartheta_{\mathcal{C}}(a_R) \leq \vartheta_{\mathcal{C}}(1-p) = 1 - \lambda_{\mathcal{C}}(p)$. If $p > \Psi_{\mathcal{C}}(q) > 1/2$, then $\lambda_{\mathcal{C}}(p) > q$, so $p_{\mathcal{C}}(q) \leq \Psi_{\mathcal{C}}(q)$. □

S2.2 One end is not enough

No noisy level below $\Psi_{\mathcal{C}}(q)$ suffices, even for log-concave W; with Lemma S3 this gives $p_{\mathcal{C}}(q) = \Psi_{\mathcal{C}}(q)$ whenever $\Psi_{\mathcal{C}}$ is convex. Suppose that $p < \Psi_{\mathcal{C}}(q)$, take ϵ with law in $\mathcal{C}$ and $E\{\min(1, q\mathrm{e}^{-\epsilon})\} > p$, let $a = \log(1/q)$, and take $\eta > 0$ so small that $E\{\min(1, q\mathrm{e}^{-\eta-\epsilon})\} > p$, which is possible by dominated convergence. Let $Y = a + \eta - X_1$, so that $F_Y(0) = q\mathrm{e}^{-\eta} < q$ and $\mathrm{pr}(Y + \epsilon \leq 0) = E\{\min(1, q\mathrm{e}^{-\eta-\epsilon})\}$, and for $s > 0$ let $W = t + sY$, which is log-concave, with noise $s\epsilon$. Then $\mathrm{pr}(|W| \leq t) \leq \mathrm{pr}(W \leq t) = q\mathrm{e}^{-\eta} < q$, while

$$\mathrm{pr}(|W + s\epsilon| \leq t) = \mathrm{pr}(Y + \epsilon \leq 0) - \mathrm{pr}(Y + \epsilon < -2t/s) \to E\{\min(1, q\mathrm{e}^{-\eta-\epsilon})\},$$

which exceeds p, as $s \to 0$. So noisy coverage p does not guarantee latent coverage q. The construction has only one end of the interval in play: the

left end is $2t/s$ tail scales away.

S2.3 Symmetric unimodal noise

By the theorem of Khintchine (Dharmadhikari and Joag-dev, 1988, Chapter 1), a law symmetric and unimodal about 0 is that of RU, with U uniform on $[-1,1]$ and $R \geq 0$ independent of U. Hence $E\{\min(1, q\mathrm{e}^{-RU})\} = E\{g(R)\} \leq \sup_r g(r)$, with $g(r) = E\{\min(1, q\mathrm{e}^{-rU})\}$, and uniform laws attain the supremum. With $\ell = \log q < 0$,

$$g(r) = q\,\frac{\sinh r}{r}\ (r \leq -\ell), \qquad g(r) = \frac{\ell + r + 1 - q\mathrm{e}^{-r}}{2r}\ (r \geq -\ell).$$

The first branch increases. On the second, $g'(r)$ has the sign of $d(r) = q\mathrm{e}^{-r}(1+r) - (1+\ell)$, which decreases, since $d'(r) = -qr\mathrm{e}^{-r}$. If $q \leq \mathrm{e}^{-1}$, then $1+\ell \leq 0 < d(r)$, so g increases throughout and $\Psi_{\mathcal{C}}(q) = \lim_{r\to\infty} g(r) = 1/2$, not attained. If $q > \mathrm{e}^{-1}$, then $d(-\ell) = \mathrm{e}^{2\ell}(1-\ell) - 1 - \ell > 0$, because this function of ℓ vanishes with its derivative at 0 and has second derivative $-4\ell\mathrm{e}^{2\ell} > 0$ for $\ell < 0$, and $d(r) \to -(1+\ell) < 0$; so g is maximal at the root r of d, where $g(r) = (1 + q\mathrm{e}^{-r})/2$. This proves the formula of Theorem 1(i) and shows that, as for Gaussian noise, $\Psi_{\mathcal{C}}(q) = 1/2$ for $q \leq \mathrm{e}^{-1}$. Heavy tails are allowed: the t_3 and Cauchy laws are symmetric and unimodal.

Proposition S1. *For symmetric unimodal noise, $\Psi = \Psi_{\mathcal{C}}$ is increasing and strictly convex on $(\mathrm{e}^{-1}, 1)$, with $\Psi(q) \to 1/2$ as $q \downarrow \mathrm{e}^{-1}$ and $\Psi(q) \to 1$*

as $q \to 1$. *Hence* $p_{\mathcal{C}}(q) = \Psi(q)$ *for* $\mathrm{e}^{-1} < q < 1$, *and noisy coverage* $p \in (1/2, 1)$ *guarantees latent coverage* $\lambda_{\mathcal{C}}(p) = \Psi^{-1}(p)$ *for every* $W \in \mathcal{B}$ *and every symmetric unimodal noise, possibly different between units; no larger latent level is guaranteed, even for log-concave* W.

Proof. Write $s = q\mathrm{e}^{-r}$ with r the root of d, so that $\Psi(q) = (1+s)/2$ and $s(1+r) = 1+\ell$. Since $\partial d/\partial r < 0$, r is differentiable in q by the implicit function theorem; differentiating $s(1+r) = 1+\ell$ with $ds = s(dq/q - dr)$ gives $rs\,dr/dq = \ell/q$ and, using $\ell = s(1+r)-1$, $ds/dq = (rs-\ell)/(qr) = (1-s)/(qr)$. Hence $\Psi'(q) = (1-s)/(2qr) > 0$ and, after the same substitution,

$$\Psi''(q) = \frac{(1-s)\{1-s(1+r)^2\}}{2q^2r^3s}, \qquad 1-s(1+r)^2 = 1-(1+\ell)(1+r).$$

Let $x = 1+\ell \in (0,1)$ and $R = 1/x - 1$. As d decreases on $(0,\infty)$, $r < R$ if and only if $d(R) < 0$, that is $\ell - R < 2\log x$, that is $k(x) = 2\log x - x + 1/x > 0$; and $k(1) = 0$, $k'(x) = -(1-x)^2/x^2 < 0$. So $(1+\ell)(1+r) < 1$ and $\Psi'' > 0$. As $q \to 1$, $0 < r < R \to 0$ and $s \to 1$, so $\Psi(q) \to 1$; as $q \downarrow \mathrm{e}^{-1}$, $x \to 0$, the root r tends to infinity and $s = (1+\ell)/(1+r) \to 0$, so $\Psi(q) \to 1/2$. Lemma S3 with $q_0 = \mathrm{e}^{-1}$ gives the guarantee and $p_{\mathcal{C}}(q) \le \Psi(q)$; § S2.2, with ϵ uniform on $[-r, r]$, gives the converse, and the same construction with $q' > \lambda_{\mathcal{C}}(p)$, for which $\Psi(q') > p$, shows that $\lambda_{\mathcal{C}}(p)$ cannot be raised. □

At $q = 0.8$, 0.9 and 0.95 the maximizing uniform half-widths are

$r = 0.26226$, 0.11332 and 0.05311, and the levels are $\Psi(q) = 0.8077245$, 0.9017873 and 0.9504311 (ball arithmetic, enclosures of width below 10^{-15}); noisy coverage 0.9 guarantees exactly latent coverage $\lambda_{\mathcal{C}}(0.9) = 0.8981432$. A certificate of the splits of § S5, run for this class as an independent check, gives $p_{\mathcal{C}} \leq 0.8078$, 0.9018 and 0.9505 at $q = 0.8$, 0.9 and 0.95, in agreement with Proposition S1.

S2.4 Mean-zero log-concave noise

Reduction. Let $h(y) = \min(1, q\mathrm{e}^{-y})$, which is continuous, nonincreasing and 1-Lipschitz. If ϵ is log-concave with mean zero, conditioning on $|\epsilon| \leq M$ and recentring changes $E\{h(\epsilon)\}$ by at most $2\mathrm{pr}(|\epsilon| > M) + |E(\epsilon \mid |\epsilon| \leq M)|$, which tends to zero; so it suffices to bound $E\{h(\epsilon)\}$ over log-concave laws on compact intervals with $E(\epsilon) \geq 0$, a larger class. By the localization theorem (Fradelizi and Guédon, 2004), the supremum of this linear functional over that class is attained at a point mass or at a law with density proportional to $\mathrm{e}^{\beta y}$ on a segment. A law with $E(\epsilon) > 0$ can be translated to mean zero without decreasing $E\{h(\epsilon)\}$, because h is nonincreasing. So the supremum is over the point mass at 0, where $h(0) = q$, and the mean-zero log-affine laws on segments. Write the latter as $\epsilon = L(V - \mu_k)$, with V on $[0, 1]$ with density proportional to e^{kv}, $k = \beta L$ and $\mu_k = E(V)$; then, with $\ell = \log q$

and $Z_k = \int_0^1 \mathrm{e}^{kv}\,dv$,

$$E\{h(\epsilon)\} = \frac{1}{Z_k}\Big\{ \int_0^{v_c} \mathrm{e}^{kv}\,dv + q\mathrm{e}^{L\mu_k} \int_{v_c}^1 \mathrm{e}^{(k-L)v}\,dv \Big\},$$

with $v_c = \min\{1, \max(0, \mu_k + \ell/L)\}$. For $|k| \geq 2$ we use the scale $\sigma = L/|k|$ and the truncated exponential form $\epsilon = \pm\sigma(\tau - m)$, with τ standard exponential truncated to $[0, |k|]$ and $m = E(\tau)$, which includes $|k| = \infty$, the centred exponential laws $\pm\sigma(X_1 - 1)$. As $|k| \geq 2$, m lies between $m_2 = 1 - 2/(\mathrm{e}^2 - 1)$ and 1.

Bounds. A branch and bound in ball arithmetic bounds $E\{h(\epsilon)\}$ from above on boxes of (k, L) for $|k| \leq 2$ and of $(1/|k|, \sigma)$ for $|k| \geq 2$, splitting until each box is below the target. On a box the bound is $E\{h(\epsilon)\} \leq \mathrm{pr}(\epsilon \leq u) + qE\{\mathrm{e}^{-\epsilon}; \epsilon > u\}$, which holds for every cut u because $h(y) \leq 1(y \leq u) + q\mathrm{e}^{-y}1(y > u)$; the cut is fixed at the kink of h for the centre of the box, so that the bound is tight to second order. The ends of the scale parameter are closed analytically.

1. For small L, $E\{h(\epsilon)\} \leq qE(\mathrm{e}^{-\epsilon}) \leq q\mathrm{e}^{L^2/24}$: with $K(s) = \log \int_0^1 \mathrm{e}^{sv}\,dv$, $\log E(\mathrm{e}^{-\epsilon}) = K(k-L) - K(k) + LK'(k) = \int_{k-L}^k (s - k + L)K''(s)\,ds \leq L^2/24$, because $K''(s) = s^{-2} - \{2\sinh(s/2)\}^{-2}$, the variance of the law on $[0, 1]$ with density proportional to e^{sv}, is at most its value $1/12$ at $s = 0$.

2. For large L, $E\{h(\epsilon)\} \le \mathrm{pr}(\epsilon \le 4) + q\mathrm{e}^{-4} \le 1 - \mathrm{e}^{-1} + 4d/L + q\mathrm{e}^{-4}$, by fact 4 of § S1 applied to V and because the density of V is at most $d = 2/(1 - \mathrm{e}^{-2})$ for $|k| \le 2$; this is below the target once L exceeds a threshold found in ball arithmetic.

3. For the truncated exponential form, with $\mathrm{pr}(\tau \le y) \le (1 - \mathrm{e}^{-y})/(1 - \mathrm{e}^{-2})$ and $\mathrm{pr}(\tau \ge y) \le \mathrm{e}^{-y}/(1 - \mathrm{e}^{-2})$: $E\{h(\epsilon)\} \le q\mathrm{e}^{\sigma}$ for $\epsilon = \sigma(\tau - m)$ and $E\{h(\epsilon)\} \le q\mathrm{e}^{-\sigma m_2}/(1-\sigma)$ for $\epsilon = \sigma(m-\tau)$ and $\sigma < 1$, at the small end, and $E\{h(\epsilon)\} \le \mathrm{pr}(\tau \le 1+4/\sigma)+q\mathrm{e}^{-4}$ and $\mathrm{pr}(\tau \ge m_2-4/\sigma)+q\mathrm{e}^{-4}$ at the large end.

Results. At $q = 0.9$ the certificate gives $\Psi_{\mathcal{C}}(0.9) < 0.9052$, and the centred exponential law $\sigma(X_1 - 1)$ with $\sigma = 0.11777$ gives $\Psi_{\mathcal{C}}(0.9) > 0.90517$; the value at that law is 0.9051755. For the two-sided level, a partition of the noisy failure probability $a_L \in [0, (1 - p)/2]$ at the left end is proposed from the values at the centred exponential law; on each interval $[a_0, a_1]$ the bound $\vartheta_{\mathcal{C}}(a_1) + \vartheta_{\mathcal{C}}(1 - p - a_0)$ is evaluated at certified lower bounds of $\lambda_{\mathcal{C}}$, obtained by the same branch and bound. This certifies the noisy level 0.906 for latent level 0.9, with 41 intervals, and noisy level 0.9 for latent level 0.8935, with 45 intervals. Numerically the worst split puts the whole failure probability at one end, as for Gaussian and symmetric unimodal noise, so the two-sided level would be $\Psi_{\mathcal{C}}(0.9) = 0.905176$; this is not used.

An open case. That the centred exponential law attains $\Psi_{\mathcal{C}}$ is supported by the certificate but not proved. Laws bounded above by $c = -\ell$ are covered analytically: if $X = -\epsilon \le s$ almost surely with $0 < s \le c$, the log-density of X minus the affine log-density of $s(1 - X_1)$ is concave on $(-\infty, s]$, so the difference of densities changes sign at most twice, and with equal means X is smaller than $s(1 - X_1)$ in the convex order; as $\Psi_{\mathcal{C}}(q) = q \sup E\{\mathrm{e}^{\min(X,c)}\}$ and $x \mapsto \mathrm{e}^{\min(x,c)}$ is convex on $(-\infty, s]$, such laws do no better than $s(1 - X_1)$. The extremal law has mass above c: at $q = 0.9$, $\sigma = 0.11777 > c = 0.10536$.

S2.5 Mean-zero unimodal noise

Let ϵ have density $(1-\rho)/a$ on $[-a, 0]$ and ρ/b on $(0, b]$, with $b = (1-\rho)a/\rho$ and $0 < \rho \le 1/2$; it is unimodal with mean zero. Since $h(y) = 1$ for $y \le \ell = \log q$, $E\{h(\epsilon)\} \ge \mathrm{pr}(\epsilon \le \ell) = (1 - \rho)(1 + \ell/a)$, which tends to one as $\rho \to 0$ and $a \to \infty$. So $\Psi_{\mathcal{C}}(q) = 1$, and no noisy level below one transfers to any latent level $q \in (0, 1)$.

S2.6 Gaussian noise: proof of Theorem 2

Let $a = \log(1/q)$, $m(b) = \phi(b)/\Phi(b)$ and $A(b) = m(b)\{b + m(b)\}$. For Z standard normal, $E(Z \mid Z \le b) = -m(b)$ and $E(Z^2 \mid Z \le b) = 1 - bm(b)$,

so $A(b) = 1 - \mathrm{var}(Z \mid Z \le b)$.

Lemma S4. *(a)* $m'(b) = -A(b)$. *(b)* $0 < A(b) < 1$ *for every real* b, $A(b) \to 1$ *as* $b \to -\infty$ *and* $A(b) \to 0$ *as* $b \to \infty$. *(c)* A *is strictly decreasing.*

Proof. (a) is direct differentiation. In (b), $A < 1$ because a truncated normal law has positive variance, and $A > 0$ because $m(b) > 0$ and $b + m(b) = \Phi(b)^{-1} \int_{-\infty}^{b} \Phi(z)\, dz > 0$; the last identity holds since both $\phi(b) + b\Phi(b)$ and $\int_{-\infty}^{b} \Phi$ have derivative $\Phi(b)$ and vanish at $-\infty$. As $b \to \infty$, $m(b) \to 0$ and $bm(b) \to 0$; as $b \to -\infty$, $m(b) = |b| + |b|^{-1} + O(|b|^{-3})$, so $A(b) \to 1$. Part (c) is inequality (4) of Sampford (1953): in his notation $\nu(x) = \phi(x)/\{1 - \Phi(x)\}$ and $\lambda(x) = \nu(x)\{\nu(x) - x\}$, so that $m(b) = \nu(-b)$ and $A(b) = \lambda(-b)$, and he proves that λ is strictly increasing on the real line. □

The derivative in s. Let $b = a/s - s$. Since $(a/s)^2 - (a/s - s)^2 = 2a - s^2$,

$$q\mathrm{e}^{s^2/2}\phi(a/s - s) = \phi(a/s). \tag{S2.1}$$

Differentiating $\Gamma(q, s) = \Phi(-a/s) + q\mathrm{e}^{s^2/2}\Phi(b)$, with $db/ds = -a/s^2 - 1$, and using (S2.1),

$$\begin{aligned}\partial_s \Gamma &= \phi(a/s)\frac{a}{s^2} + q\mathrm{e}^{s^2/2}\Big\{s\Phi(b) - \phi(b)\Big(\frac{a}{s^2} + 1\Big)\Big\} \\ &= qs\mathrm{e}^{s^2/2}\Phi(b) - \phi(a/s) = \phi(a/s)\Big\{\frac{s}{m(b)} - 1\Big\},\end{aligned}$$

since $q\mathrm{e}^{s^2/2}\Phi(b) = q\mathrm{e}^{s^2/2}\phi(b)/m(b) = \phi(a/s)/m(b)$. So $\partial_s\Gamma$ has the sign of $s - m(b)$.

The sign change. For fixed $a > 0$, $s \mapsto b = a/s - s$ is a strictly decreasing bijection of $(0, \infty)$ onto $\mathbb{R}$, and $s^2 + bs = a$. The map $t \mapsto t^2 + bt$ is strictly increasing on $t > -b/2$, and both s and $m(b)$ lie there: $s = \{-b + (b^2 + 4a)^{1/2}\}/2 > -b/2$, and $m(b) > \max(0, -b) \geq -b/2$ by Lemma S4(b). Hence

$$s > m(b) \iff a = s^2 + bs > m(b)^2 + bm(b) = A(b).$$

As s increases, b decreases, and $A(b)$ increases strictly through all of $(0, 1)$ by Lemma S4.

Part (i). If $a \geq 1$, that is $q \leq \mathrm{e}^{-1}$, then $a > A(b)$ for every s, so $\Gamma(q, \cdot)$ is strictly increasing. As $s \to \infty$, $\Phi(-a/s) \to 1/2$ and $q\mathrm{e}^{s^2/2}\Phi(b) = \phi(a/s)/m(b) \to 0$, because $b \to -\infty$ and $m(b) \to \infty$. So $\Psi_{\mathcal{N}}(q) = 1/2$, approached as $s \to \infty$ and not attained.

Part (ii). If $a < 1$, $\partial_s\Gamma$ is positive before and negative after the unique s^* with $A(b^*) = a$, so s^* is the unique maximizer, and $s^* = m(b^*)$ there. At s^*, $a/s^* = b^* + s^* = c$, and $q\mathrm{e}^{s^{*2}/2}\Phi(b^*) = \phi(c)/m(b^*)$, which gives the closed form.

Part (iii): the derivative in q. With s fixed, by (S2.1),

$$\partial_a G = -\frac{\phi(a/s)}{s} - q\mathrm{e}^{s^2/2}\Phi(b) + q\mathrm{e}^{s^2/2}\frac{\phi(b)}{s} = -q\mathrm{e}^{s^2/2}\Phi(b),$$

so $\partial_q \Gamma = -q^{-1}\partial_a G = \mathrm{e}^{s^2/2}\Phi(b)$. The maximizer $s^*(q) = m\{A^{-1}(\log 1/q)\}$ is continuous in q, as A is a continuous strictly monotone bijection of $\mathbb{R}$ onto $(0,1)$. For $q_1 < q_2$ in $(\mathrm{e}^{-1}, 1)$, with $s_j = s^*(q_j)$,

$$\Gamma(q_2, s_1) - \Gamma(q_1, s_1) \le \Psi_{\mathcal{N}}(q_2) - \Psi_{\mathcal{N}}(q_1) \le \Gamma(q_2, s_2) - \Gamma(q_1, s_2);$$

dividing by $q_2 - q_1$ and using the continuity of s^* and of $\partial_q\Gamma$ in both arguments gives $\Psi'_{\mathcal{N}}(q) = \partial_q\Gamma\{q, s^*(q)\} = D(b^*)$ with $D(b) = \mathrm{e}^{m(b)^2/2}\Phi(b)$, which is continuous in q.

Monotonicity. Along the maximizers, $q(b) = \mathrm{e}^{-A(b)}$ increases strictly from e^{-1} to 1 as b runs over $\mathbb{R}$, by Lemma S4. By Lemma S4(a),

$$\frac{d}{db}\log D(b) = m(b)m'(b) + m(b) = m(b)\{1 - A(b)\} = m(b)\mathrm{var}(Z \mid Z \le b) > 0,$$

so $\Psi'_{\mathcal{N}}$ is strictly increasing on $(\mathrm{e}^{-1}, 1)$ and $\Psi_{\mathcal{N}}$ is strictly convex there. As $q \downarrow \mathrm{e}^{-1}$, $b^* \to -\infty$; then $c = b^* + m(b^*) \to 0$, so $\Psi_{\mathcal{N}}(q) \to 1/2$, and $m^2 = b^2 + 2 + o(1)$ and $\Phi(b) = \phi(b)\{1 + o(1)\}/|b|$, so $D(b) = \mathrm{e}\{1 + o(1)\}/\{(2\pi)^{1/2}|b|\} \to 0$, the derivative of the constant part. As $\Psi_{\mathcal{N}}$ is continuous at e^{-1}, the mean value theorem then gives a right derivative 0 there, matching the left one. Hence $\Psi_{\mathcal{N}}$ is continuously differentiable and convex on $(0,1)$. As $q \to 1$, $b^* \to \infty$, $c \to \infty$ and $\Psi_{\mathcal{N}}(q) \to 1$.

The two-sided level. Lemma S3 with $q_0 = \mathrm{e}^{-1}$ gives $p_{\mathcal{N}}(q) \le \Psi_{\mathcal{N}}(q)$, and § S2.2 gives $p_{\mathcal{N}}(q) \ge \Psi_{\mathcal{N}}(q)$. This completes the proof. □

The proof uses only Lemma S4. What it adds is the reduction: the first-order condition is $s = m(b)$, a Mills ratio of the noise; the envelope derivative is a tilted distribution function, $\mathrm{e}^{s^2/2}\Phi(b)$; and its monotonicity along the maximizers is the monotonicity of a truncated variance. It also explains the threshold: $\mathrm{e}^{-1} = \mathrm{e}^{-\sup A}$. Table S3 gives the constants at three levels; the enclosures of $\Psi_{\mathcal{N}}(q)$ have width below 3×10^{-16} (§ S5).

Table S3: Constants of Theorem 2 and of Proposition S1

	Gaussian noise					symmetric unimodal noise	
q	b^*	$s^* = m(b^*)$	c	$\Psi_{\mathcal{N}}(q)$	$\Psi'_{\mathcal{N}}(q)$	r	$\Psi_{\mathcal{C}}(q)$
0.8	1.51627	0.135124	1.65139	0.8044106	0.94385	0.26226	0.8077245
0.9	2.04546	0.050274	2.09573	0.9008035	0.98083	0.11332	0.9017873
0.95	2.43139	0.020916	2.45231	0.9501590	0.99270	0.05311	0.9504311

s^* is the maximizing noise scale in units of the tail scale of the extremal exponential tail; r is the maximizing half-width of the uniform noise law in the same units. The inverse levels are $\lambda_{\mathcal{N}}(0.8) = 0.79532$, $\lambda_{\mathcal{N}}(0.9) = 0.89918$ and $\lambda_{\mathcal{N}}(0.95) = 0.94984$.

S2.7 Unimodal latent laws: proof of Proposition 1

The bound needs no shape. Given $W > 1$, the event $\epsilon \ge 0$ gives $W + \epsilon > 1$, and given $W < -1$ the event $\epsilon \le 0$ gives $W + \epsilon < -1$; both have probability at least $\beta_{\mathcal{C}}$, since $\mathcal{C}$ is closed under negation and ϵ is independent of W, so $1 - p \ge \mathrm{pr}(|W + \epsilon| > 1) \ge \beta_{\mathcal{C}}\mathrm{pr}(|W| > 1)$. For median-zero noise this is the bound of Guille-Escuret and Ndiaye (2024); for mean-zero log-concave noise

$\mathrm{pr}(\epsilon \geq 0) \geq \mathrm{e}^{-1}$ and $\mathrm{pr}(\epsilon \leq 0) \geq \mathrm{e}^{-1}$ by fact 4 of § S1, with equality for the centred exponential law $X_1 - 1$; for mean-zero unimodal noise $\beta_{\mathcal{C}} = 0$, by the laws of § S2.5, for which $\mathrm{pr}(\epsilon \geq 0) = \rho$.

For the construction, let ϵ have a law in $\mathcal{C}$, continuous at 0, with $\mathrm{pr}(\epsilon \geq 0) = \beta < \beta_{\mathcal{C}} + \gamma'$; as $p > 1 - \beta_{\mathcal{C}}$, $(1-p)/\beta < 1$ for small γ'. Let $\tau, \gamma'' > 0$ be small and $0 < \omega < 1$, with ω to tend to 0 with the noise scale. Let

$$s = (1 - p - \gamma'' - \tau)/\beta, \qquad m = 1 - s - \tau, \qquad A = 2s(1 + \omega/2)/\tau - 1,$$

so that $s, m > 0$, and let W have a density made of three pieces: $m/2$ on $[-1, 1]$, the mass inside the interval; s/ω on $(1, 1 + \omega]$, a thin layer just outside the right end; and $\tau/(A-1)$ on $[-A, -1)$, a long light left tail that makes $E(W) = s(1 + \omega/2) - \tau(A+1)/2 = 0$. For small τ and ω the density is nondecreasing up to $1 + \omega$ and zero beyond it, so W is unimodal. With noise $\sigma\epsilon$, the failure probability of $|W + \sigma\epsilon| \leq 1$ splits over the three pieces: the inside piece contributes $(m/2) \int_{-1}^{1} \mathrm{pr}(|w + \sigma\epsilon| > 1)\, dw$, which tends to 0 as $\sigma \to 0$ by dominated convergence; the layer contributes at most $s\{\mathrm{pr}(\epsilon > -\omega/\sigma) + \mathrm{pr}(\epsilon < -2/\sigma)\}$, where the second term tends to 0 and the first tends to β as $\sigma \to 0$ with $\omega/\sigma \to 0$, for instance $\omega = \sigma^2$, by continuity at 0; and the left tail contributes at most τ. So the noisy coverage tends to at least $1 - \tau - s\beta = p + \gamma'' > p$, while the latent coverage is $m = 1 - (1 - p - \gamma'' - \tau)/\beta - \tau$, within any γ of $1 - (1-p)/\beta_{\mathcal{C}}$ for small

γ', γ'' and τ. If $\beta_{\mathcal{C}} = 0$, take $\beta < \gamma'$ and $s = 1 - \tau - \gamma$, so that the latent coverage is $m = \gamma$ while the noisy coverage tends to at least $1 - \tau - s\beta > p$ for small τ and γ'.

The noisy coverage of this law has a closed form: for a piece with uniform density on $[a, b]$, $\int_a^b F(c - w)\, dw = \bar{G}(c - a) - \bar{G}(c - b)$, where F is the distribution function of the noise and $\bar{G}$ an antiderivative, $\bar{G}(y) = y\Phi(y/\sigma) + \sigma\phi(y/\sigma)$ for Gaussian noise. With $\tau = 10^{-4}$, a layer of width $\omega = 10^{-3}\sigma$, a thousandth of a noise standard deviation, and the noisy coverage set to 0.9, the latent coverage is 0.870, 0.820, 0.807 and 0.801 at $\sigma = 0.1$, 0.03, 0.01 and 0.001 for Gaussian noise, 0.876, 0.822, 0.807 and 0.801 for uniform noise of standard deviation σ, and 0.809, 0.751, 0.736 and 0.729 for the centred exponential noise $\sigma(X_1 - 1)$, against the limits 0.8, 0.8 and $1 - \mathrm{e}/10 = 0.72817$.

S3. Calibration with unknown, heterogeneous noise

This section completes the proofs of § 4.

S3.1 Theorem 3 and the ranks of Table 2

The converse in Theorem 3. Let $p < \Psi_{\mathcal{C}}(q)$ and take ϵ, η and $Y = a + \eta - X_1$ as in § S2.2. For $s > 0$ let every unit have the log-concave latent law G of

$W = 1 + sY$ and the noise $s\epsilon$. Then $\mathrm{pr}(|W| \le 1) < q$, so $r = Q_q(|W|) > 1$, and $\mathrm{pr}(|V_i| < r) \ge \mathrm{pr}(|V_i| \le 1) > p$ for small s. The indicators $1(|V_i| < r)$ are independent with a common success probability above p, so the latent reliability $\mathrm{pr}(N \le k-1)$ is at most $\mathrm{pr}\{\mathrm{Bin}(K,p) \le k-1\}$. Letting $p \uparrow \Psi_{\mathcal{C}}(q)$ gives the claim, since the binomial probability is continuous in p.

Sharpness. Let $\ell \in (0,1)$ and let k_K be the smallest rank with $\mathrm{pr}\{\mathrm{Bin}(K,\ell) \le k_K - 1\} \ge 1 - \delta$. By minimality,

$$1 - \delta \le \mathrm{pr}\{\mathrm{Bin}(K,\ell) \le k_K - 1\} < 1 - \delta + \mathrm{pr}\{\mathrm{Bin}(K,\ell) = k_K - 1\}$$
$$\le 1 - \delta + \max_j \mathrm{pr}\{\mathrm{Bin}(K,\ell) = j\},$$

and the largest binomial probability is of order $K^{-1/2}$. With $\ell = \Psi_{\mathcal{C}}(q)$, the infimum over the laws of the converse of the latent reliability of the rank k_K is $\mathrm{pr}\{\mathrm{Bin}(K, \Psi_{\mathcal{C}}(q)) \le k_K - 1\}$, which tends to $1 - \delta$; the exact values in § S6.1 show the approach from above. This makes the rank sharp where the valid rank is computed from $p_{\mathcal{C}}(q) = \Psi_{\mathcal{C}}(q)$, that is for Gaussian and symmetric unimodal noise. For mean-zero log-concave noise the rank is computed from the certified sufficient level, which exceeds the certified lower bound on $\Psi_{\mathcal{C}}(q)$; at the centred exponential law its reliability then tends to one, not to $1 - \delta$, and the smallest valid rank is known only to lie in the range of Table 2.

Symmetric unimodal latent laws: the first row of Table 2. If W is

symmetric and unimodal about 0, then for every real c the interval $[c - s, c + s]$ carries at most the mass of $[-s, s]$, so $\mathrm{pr}(|W + e| \le s \mid e) \le \mathrm{pr}(|W| \le s)$ for any noise e independent of W, and integrating over e gives $\mathrm{pr}(|V_i| \le s) \le \mathrm{pr}(|W| \le s)$ for every unit, whatever its noise law and scale. For $s < r = Q_q(|W|)$ this is below q, so $\mathrm{pr}(|V_i| < r) \le q$ and the proof of Theorem 3 gives the usual rank, level q, for any independent and heterogeneous noise; it cannot be lowered, since noise-free scores are allowed. This is the one-dimensional case of Anderson (1955).

When a valid rank exists. A rank $k \le K$ with $\mathrm{pr}\{\mathrm{Bin}(K, \ell) \le k - 1\} \ge 1 - \delta$ exists if and only if $1 - \ell^K \ge 1 - \delta$, that is $K \ge \log \delta / \log \ell$. At $q = 0.9$ and $\delta = 0.05$ this gives $K \ge 29$ for the Gaussian and symmetric unimodal levels 0.9008035 and 0.9017873, $K \ge 31$ for the sufficient log-concave level 0.906, and $K \ge 59$ and $K \ge 80$ for the levels 0.95 and $1 - 0.1/\mathrm{e}$ without a latent shape assumption. All ranks in the main paper are computed from the exact binomial distribution function, with ties within 10^{-10} of $1 - \delta$ rechecked at 50 digits.

Unimodal latent laws: the last two rows of Table 2. The proof of Theorem 3 uses only the transfer for one unit. With the transfer of Proposition 1, which needs noisy level $1 - \beta_{\mathcal{C}}(1 - q)$, that is $(1 + q)/2$ for median-zero noise and $1 - (1 - q)/\mathrm{e}$ for mean-zero log-concave noise, it gives the ranks of the

last two rows: if $\mathrm{pr}(|W| \le s) < q$ then $\mathrm{pr}(|V_i| \le s) < 1-\beta_{\mathcal{C}}(1-q)$, for every latent law. These ranks are the smallest valid for unimodal latent laws with mean zero. Let $p < 1-\beta_{\mathcal{C}}(1-q)$, and take the construction of § S2.7 with the layer mass s chosen in $1-q-\tau < s < (1-\tau-p)/\beta$, which is possible for small τ and γ'. The latent coverage of $[-1,1]$ is then $m = 1-s-\tau < q$, so $r = Q_q(|W|) > 1$, and $\mathrm{pr}(|V_i| < r) \ge \mathrm{pr}(|V_i| \le 1) > p$ for small σ; as above, the reliability of rank k is at most $\mathrm{pr}\{\mathrm{Bin}(K,p) \le k-1\}$.

One-sided intervals. For $(-\infty, T]$ with $T = V_{(k)}$, the kth smallest signed score, let $r = F_W^{-1}(q)$ and $N = \#\{i : V_i < r\}$; then $\mathrm{pr}(W_{\mathrm{new}} \le T \mid \mathcal{D}) \ge q$ if and only if $N \le k-1$. By Proposition 2, applied to $Y = (W-s)/\sigma_i$ for $s < r$, $\mathrm{pr}(V_i \le s) \le \Psi_{\mathcal{C}}(q)$, so $\mathrm{pr}(V_i < r) \le \Psi_{\mathcal{C}}(q)$, which needs only concavity of $\log F_W$; the converse of Proposition 2 gives the matching lower bound. The smallest valid rank is therefore the smallest k with $\mathrm{pr}\{\mathrm{Bin}(K, \Psi_{\mathcal{C}}(q)) \le k-1\} \ge 1-\delta$, for every noise class with $\beta_{\mathcal{C}} > 0$: at $q = 0.9$ the level is 0.9008035 for Gaussian noise, 0.9017873 for symmetric unimodal noise and in $(0.90517, 0.9052)$ for mean-zero log-concave noise. *Asymmetric intervals.* Bi-log-concavity allows skewed latent laws, for which an interval $[-T_L, T_R]$ may be shorter than $[-T, T]$. Split the latent failure as $\beta_L + \beta_R = 1-q$, fixed in advance, and let T_R be the order statistic of the signed scores V_i at the one-sided rank for latent level $1-\beta_R$ and reliability $1-\delta/2$, and T_L

that of the $-V_i$ for $1-\beta_L$. The right end uses concavity of $\log F_W$ and the left end concavity of $\log(1-F_W)$, so for $W \in \mathcal{B}$ both ends hold, each with probability at least $1-\delta/2$, and the union bound gives latent coverage q with probability at least $1-\delta$; each end is exact, and only the union bound and the choice of split are conservative. We have not studied the widths of this rule.

Without a shape assumption the one-sided level is $1-\beta_{\mathcal{C}}(1-q)$: it suffices because $W > t$ and $e \geq 0$ give $W + e > t$, and a layer of mass slightly above $1-q$ just above t, with noise whose probability of being nonnegative is close to $\beta_{\mathcal{C}}$, has latent coverage below q and noisy coverage close to $1-\beta_{\mathcal{C}}(1-q)$.

Proposition 3. The proof in the main text uses only that $\Psi_{\mathcal{N}}(q) > q$ for every $q \in (0,1)$: for $q \leq \mathrm{e}^{-1}$, $\Psi_{\mathcal{N}}(q) = 1/2 > q$; for $q > \mathrm{e}^{-1}$, $\Gamma(q,s) \to q$ as $s \to 0$ and $\Gamma(q,\cdot)$ increases near 0, by the sign of $\partial_s \Gamma$ in § S2.6. The converse construction above, with Gaussian noise and $q < p < \Psi_{\mathcal{N}}(q)$, then gives $\mathrm{pr}(|V_i| < r) > p$, and the law of large numbers does the rest. If $k_K = Kq + z\{Kq(1-q)\}^{1/2} + O(1)$, as for the usual rank with z the standard normal $(1-\delta)$-quantile, and $H = \mathrm{pr}(|V_i| < r)$, the normal approximation gives the latent reliability $\Phi[z - K^{1/2}(H-q)/\{q(1-q)\}^{1/2}]$, which is $1/2$ at $K = z^2 q(1-q)/(H-q)^2$; with $H = \Psi_{\mathcal{N}}(q)$ this is 2.2×10^4, 3.8×10^5

and 5.1×10^6 at $q = 0.8$, 0.9 and 0.95, close to the exact crossings below 0.5 found by scanning every integer K, $K = 22{,}177$, 376,330 and 5,080,045.

S3.2 Latent laws that differ between units

A general form of Corollary 1. Without the quantile condition, let $c_i = \mathrm{pr}(|W_i| < r_0)$. For $s < r_0$ and every $q' > c_i$, $\mathrm{pr}(|W_i| \le s) \le c_i < q'$, and the transfer gives $\mathrm{pr}(|V_i| \le s) \le p_{\mathcal{C}}(q')$; hence $\pi_i = \mathrm{pr}(|V_i| < r_0) \le \bar{p}_i = \inf_{q' > c_i} p_{\mathcal{C}}(q')$. The number N of the proof of Theorem 3 is a sum of independent Bernoulli variables with probabilities $\pi_i \le \bar{p}_i$, so it is stochastically smaller than the sum with probabilities $\bar{p}_i$, and by Lemma S6 below

$$\mathrm{pr}_{\mathcal{D}}\{\mathrm{pr}(|W_{\mathrm{new}}| \le T \mid \mathcal{D}) \ge q\} \ge \mathrm{pr}\{\mathrm{Bin}(K, \bar{\pi}) \le k-1\}, \qquad \bar{\pi} = K^{-1} \sum_{i=1}^{K} \bar{p}_i,$$

whenever $k \ge K\bar{\pi} + 1$, by Hoeffding (1956, Theorem 4); Lemma S6 below gives a self-contained proof when $k \ge K\bar{\pi} + 2$. Here $p_{\mathcal{C}}(q')$ is available for $q' > \mathrm{e}^{-1}$ in the Gaussian and symmetric unimodal classes; elsewhere take $\bar{p}_i = 1$. If the rank is that of Theorem 3, the bound is at least $1-\delta$ whenever $\bar{\pi} \le p_{\mathcal{C}}(q)$, which is the precise sense in which slack is absorbed. Under the condition of Corollary 1 its proof gives $\pi_i \le p_{\mathcal{C}}(q)$ directly, and the bound of Theorem 3 needs no condition on k. A few calibration units that are easier than the new one, with $\bar{p}_i > p_{\mathcal{C}}(q)$, are absorbed if the others have slack, $\bar{p}_i < p_{\mathcal{C}}(q)$. The c_i are not identifiable from the data, so this form matters

for interpretation, not for computing a rank. The extra unit in $k \geq K\bar{\pi} + 2$ is harmless: the ranks of Theorem 3 exceed $K\bar{\pi}$ by about $z\{K\bar{\pi}(1-\bar{\pi})\}^{1/2}$.

Lemma S5. *Let $S = \sum_{i=1}^{n} X_i$ with independent $X_i \sim$ Bernoulli(p_i), $\mu = \sum_i p_i$ and $a_k = \mathrm{pr}(S = k)$. If $j \geq \mu$ is an integer, then $a_{j+1} \leq a_j$.*

Proof. Fix n, j and $\mu \leq j$, and let $\Delta(p) = a_{j+1} - a_j$ on the compact set $P_\mu = \{p \in [0,1]^n : \sum_i p_i = \mu\}$. Fix a pair (i, l), let U be the sum of the other $n - 2$ variables, with $u_k = \mathrm{pr}(U = k)$, and write $s = p_i + p_l$ and $\varpi = p_i p_l$. Then $a_k = (1-s)u_k + s u_{k-1} + \varpi(u_k - 2u_{k-1} + u_{k-2})$, so with s fixed $\Delta = \alpha(s) + \gamma\varpi$, where γ depends only on U. On the segment $\{p_i + p_l = s\}$, ϖ is largest at $p_i = p_l$ and smallest at the endpoints, where one of the two is 0 or 1.

Among the maximizers of Δ on P_μ, a nonempty closed set, take one with the smallest $\sum_i p_i^2$. Suppose that two of its coordinates lie in $(0, 1)$ and are unequal. If $\gamma < 0$, moving to an endpoint of the segment strictly increases Δ, contradicting maximality; if $\gamma \geq 0$, setting both to $s/2$ does not decrease Δ and strictly decreases $\sum_i p_i^2$, contradicting the choice. So the fractional coordinates of this maximizer are all equal, to α say, and $S = m + B$, with m coordinates equal to 1, $B \sim \mathrm{Bin}(r, \alpha)$, and $m + r\alpha = \mu \leq j$. Let $i = j - m \geq r\alpha \geq 0$. If $r = 0$ or $i \geq r$, then $a_{j+1} = 0$. Otherwise $a_{j+1}/a_j = (r-i)\alpha/\{(i+1)(1-\alpha)\}$, which is at most 1 because

$i \geq r\alpha > r\alpha - (1-\alpha)$. Hence $\max_{P_\mu} \Delta \leq 0$. □

Lemma S5 also follows from the theorem of Darroch (1964) on the mode of S together with the log-concavity of its probabilities; the proof above is self-contained.

Lemma S6. *In the setting of Lemma S5, let* $\bar{p} = \mu/n$. *If* c *is an integer with* $c \geq n\bar{p} + 1$, *then* $\mathrm{pr}(S \leq c) \geq \mathrm{pr}\{\mathrm{Bin}(n, \bar{p}) \leq c\}$.

Proof. Replace (p_i, p_l) by their mean and let R be the sum of the other variables. The law of $X_i + X_l$ gains $d = (p_i - p_l)^2/4 \geq 0$ at 0 and at 2 and loses $2d$ at 1, so $\mathrm{pr}(S \leq c)$ changes by $d\{F_R(c) - 2F_R(c-1) + F_R(c-2)\} = d\{\mathrm{pr}(R = c) - \mathrm{pr}(R = c-1)\} \leq 0$, by Lemma S5 for R with $j = c-1$, since the mean of R is at most $n\bar{p} \leq c-1$. Each replacement keeps $\sum_i p_i$. Averaging the largest and smallest coordinates reduces $\sum_i (p_i - \bar{p})^2$ by $(p_{\max} - p_{\min})^2/2$, at least $1/(2n)$ times its value, so iterating drives p to $(\bar{p}, \ldots, \bar{p})$; as $\mathrm{pr}(S \leq c)$ is continuous in p and never increases along the way, the limit gives the claim. □

Hoeffding (1956, Theorem 4) gives the same comparison under the weaker condition $c \geq n\bar{p}$; on 20,000 random probability vectors with $n \leq 40$ the inequality held under that condition too, with largest violation 9×10^{-16}, rounding.

Normalized scores. Let $s_i > 0$ be known scales fixed before calibration, such as functions of the training fit or of area sizes, and use the scores $|V_i|/s_i$ and the interval $[-s_{\text{new}}T, s_{\text{new}}T]$. Then $W_i/s_i \in \mathcal{B}$ and e_i/s_i has a law in $\mathcal{C}$, because both classes are closed under scaling, and the proof of Corollary 1 applies with W_i/s_i and $W_{\text{new}}/s_{\text{new}}$ in place of W_i and W_{new}. Latent laws that differ only in a known scale therefore give Theorem 3 exactly; this is the usual normalized conformal score and covers heteroscedastic latent residuals.

Why a link between units is needed. Suppose that the new unit is a random draw from a heterogeneous population, so that W_{new} follows the mixture of the calibration laws. Point masses lie in $\mathcal{B}$, and their mixtures are arbitrary laws, so Proposition 1 applies and the guarantee is the shape-free one. Concretely, let each unit be of one of two types, with latent point masses at 0 and at $1+\eta$, the second type with probability w slightly above $1-q$, and let the noise be $N(0,\sigma^2)$ with σ small. Then $r_0 = Q_q(|W_{\text{new}}|) = 1+\eta$, while $\text{pr}(|V_i| < r_0) \to (1-w) + w/2 = 1 - w/2$, about $(1+q)/2$, as $\sigma \to 0$. This exceeds $\Psi_{\mathcal{N}}(q)$, so the latent reliability of the rank of Theorem 3 tends to zero as K grows (§ S6.5), while the shape-free rank, at level $(1+q)/2$, stays valid. If instead the population law itself is bi-log-concave and the noise is independent of the type of unit, Theorem 3 applies

directly to it.

S3.3 Contaminated latent laws: proof of Proposition 5

Let $\mathcal{C}$ be the Gaussian or the symmetric unimodal class, so that $\beta = \beta_{\mathcal{C}} = 1/2$, and write $\Psi = \Psi_{\mathcal{C}}$ and $\lambda = \lambda_{\mathcal{C}}$. By Theorem 2(iii) and Proposition S1, Ψ is continuous, increasing, differentiable and strictly convex on $(\mathrm{e}^{-1}, 1)$, with $\Psi \to 1/2$ at e^{-1} and $\Psi \to 1$ at 1; hence λ is increasing, differentiable and concave on $(1/2, 1)$, with $\lambda' = 1/\Psi'(\lambda)$. Let $q_\beta \in (\mathrm{e}^{-1}, 1)$ solve $\Psi'(q_\beta) = 1/2$: $q_\beta = 0.44635$ and $\Psi(q_\beta) = 0.52634$ for Gaussian noise, $q_\beta = 0.42611$ and $\Psi(q_\beta) = 0.52267$ for symmetric unimodal noise (double precision). The condition $q \geq 0.53$ of Proposition 5 ensures $q \geq \Psi(q_\beta)$ in both classes.

Sufficiency. Fix $t > 0$, $\xi \in [0, 1-q)$, and write W as W_0 with probability $1 - \xi$ and W_H with probability ξ, through a label independent of the noise e, with $W_0 \sim G \in \mathcal{B}$ and $W_H \sim H$. Let $P_0 = \mathrm{pr}(|W_0 + e| \leq t)$ and $x = \mathrm{pr}(|W_H + e| \leq t)$, so that the noisy coverage is $P = (1 - \xi)P_0 + \xi x$. By Lemma S3, $\mathrm{pr}(|W_0| \leq t) \geq \lambda(P_0)$, with $\lambda(P_0) = 0$ for $P_0 \leq 1/2$. Without a shape assumption, $\mathrm{pr}(|W_H| \leq t) \geq f(x) = \max\{0, 1 - 2(1 - x)\}$, because $|W_H| > t$ and e of the same sign as W_H give $|W_H + e| > t$, and $\mathrm{pr}(e \geq 0), \mathrm{pr}(e \leq 0) \geq 1/2$. Hence the latent coverage is at least

$$J_P(x) = (1 - \xi)\lambda\{P_0(x)\} + \xi f(x), \qquad P_0(x) = (P - \xi x)/(1 - \xi),$$

minimized over the $x \in [0,1]$ with $P_0(x) \in [0,1]$. As λ and f are nondecreasing, this minimum is nondecreasing in P, so it suffices to show that it exceeds q for $P \in (p_\xi(q), 1-\xi/2)$; note that $p_\xi(q) < 1-\xi/2$ because $\Psi < 1$ on $(0,1)$. On $x \in [0,1/2]$, $f = 0$ and J_P is nonincreasing, so its minimum there is at $x = 1/2$, which is feasible since $P \le 1-\xi/2$. On $x \in [1/2,1]$, $P_0(x)$ runs over $[P_0(1), P_0(1/2)]$. Since $P > p_\xi(q)$, $q_0 = q/(1-\xi) < 1$ and $\Psi(q_0) \ge q_0$,

$$P_0(1) = \frac{P-\xi}{1-\xi} > \Psi(q_0) - \frac{\xi}{2(1-\xi)} \ge \frac{q-\xi/2}{1-\xi} \ge q,$$

the last step because $q \ge 1/2$. So $P_0(1) > q \ge \Psi(q_\beta) > 1/2$ and $P_0(1/2) < 1$, λ is concave on $[P_0(1), P_0(1/2)]$, J_P is concave on $[1/2,1]$, and its minimum there is at an end. By concavity, $\lambda\{P_0(1/2)\} - \lambda\{P_0(1)\} \le \lambda'\{P_0(1)\}\xi/\{2(1-\xi)\}$, and $\lambda'\{P_0(1)\} = 1/\Psi'[\lambda\{P_0(1)\}] \le 1/\Psi'(q_\beta) = 2$ because $\lambda\{P_0(1)\} > q_\beta$ and Ψ' is nondecreasing. Hence

$$J_P(1) - J_P(1/2) \ge \xi - (1-\xi)\,\lambda'\{P_0(1)\}\,\frac{\xi}{2(1-\xi)} \ge 0,$$

and the minimum of J_P is $J_P(1/2) = (1-\xi)\lambda[\{P-\xi/2\}/(1-\xi)]$, which exceeds $(1-\xi)\lambda\{\Psi(q_0)\} = q$. For $\xi \ge 1-q$, the level $(1+q)/2 = 1-\beta(1-q)$ suffices for every law, by Proposition 1.

Sharpness. Let $\xi < 1-q$, $q_0 = q/(1-\xi)$, $a_0 = \log(1/q_0)$, and let ϵ^* attain $\Psi(q_0)$: s^*Z for Gaussian noise (Theorem 2), uniform on $[-r,r]$ for

symmetric unimodal noise; in both cases $\mathrm{pr}(\epsilon^* < 0) = 1/2$ and ϵ^* has no atom. For $\eta, \eta' > 0$ and $s > 0$ let G be the law of $t + s(a_0 + \eta - X_1)$, which is log-concave, let H be the point mass at $t + s\eta'$, and let the noise be $s\epsilon^*$. The latent coverage is at most $(1-\xi)q_0 \mathrm{e}^{-\eta} < q$, since $t + s\eta' > t$. As $s \to 0$, the noisy coverage tends to $(1-\xi)E\{\min(1, q_0 \mathrm{e}^{-\eta-\epsilon^*})\} + \xi \mathrm{pr}(\epsilon^* < -\eta')$ (§ S2.2), and as $\eta, \eta' \to 0$ to $(1-\xi)\Psi(q_0) + \xi/2 = p_\xi(q)$. So no level below $p_\xi(q)$ suffices. For $\xi \geq 1 - q$, take G the point mass at 0 and H with mass w at $t + s\eta'$ and $1 - w$ at 0, with ξw slightly above $1 - q$; the latent coverage is $1 - \xi w < q$ and the noisy coverage tends to $1 - \xi w/2$, close to $(1+q)/2$.

Monotonicity in ξ. With $q_0 = q/(1-\xi)$, $\partial_\xi p_\xi(q) = 1/2 - \{\Psi(q_0) - q_0\Psi'(q_0)\}$. By convexity of Ψ on $[\mathrm{e}^{-1}, q_0]$ and $\Psi(\mathrm{e}^{-1}+) = 1/2$, the tangent at q_0 lies below Ψ at e^{-1}, so $\Psi(q_0) - q_0\Psi'(q_0) \leq 1/2 - \Psi'(q_0)/\mathrm{e} < 1/2$, and $p_\xi(q)$ increases strictly in ξ; at $\xi = 1 - q$ it equals $q + (1-q)/2 = (1+q)/2$, so it is continuous there.

The rank. The proof of Theorem 3 uses the transfer for one unit at thresholds $s < r = Q_q(|W|)$; with the transfer above, $\mathrm{pr}(|V_i| < r) \leq p_\xi(q)$ for every $W_i \in \mathcal{B}_\xi$, and the converse and sharpness follow from the laws of the sharpness part, used in every unit, as in § S3.1; Corollary 1 carries over unchanged, since $\mathcal{B}_\xi$ increases with ξ. Since $\mathrm{pr}\{\mathrm{Bin}(K, \ell) \leq k - 1\} = \mathrm{pr}\{\mathrm{Beta}(k, K+1-k) > \ell\}$ decreases in ℓ, rank k is valid over $\mathcal{B}_\xi$ if and only

if $p_\xi(q)$ is at most the δ-quantile of $\mathrm{Beta}(k, K+1-k)$, and the contamination it tolerates is the root of that equation in ξ. If calibration and new units are independent draws from a population whose latent law is in $\mathcal{B}_\xi$, and the noise is independent of W given the component label with law in $\mathcal{C}$, the argument applies to a random unit.

Scope. For $q < 0.53$ the closed form can be too low and is not claimed: when $P_0(1) \le 1/2$, the jump of λ at $1/2$ adds a candidate to the minimum of J_P; at $q = 0.4$ and $\xi = 0.3$ the closed form gives 0.5747, a direct minimization 0.65. For mean-zero log-concave noise the sufficiency argument holds with the certified transfer, but its evaluation needs convexity of the two-sided transfer, which is not established; the construction above with centred exponential noise gives the lower bound $(1 - \xi)\max_\sigma E[\min\{1, q_0 \mathrm{e}^{-\sigma(X_1-1)}\}] + \xi(1 - \mathrm{e}^{-1})$ on any valid level. The same argument with the one-sided transfer gives the same level for one-sided intervals.

S3.4 Centred residuals: proof of Proposition 4

Coverages in the definition of $p_{\mathcal{C}}^{\zeta}(q)$ are of the open interval $(-1, 1)$. For continuous noise the noisy coverage is the same on the closed interval, and a log-concave law that is not a point mass has no atom; a point mass with

$\mathrm{pr}(|W| < 1) \le q < 1$ lies at $|w_0| \ge 1$ and has noisy coverage at most $1/2$.

Part (i). Let $r = Q_q(|W|) > 0$ and $N = \#\{i : |V_i| < r\}$, so that the latent coverage is at least q if and only if $N \le k - 1$, as in the proof of Theorem 3. Since r is the lower quantile, $\mathrm{pr}(|W| < r) \le q$. The law of W/r is log-concave with $|E(W/r)| \le \zeta$ and $\mathrm{pr}(|W/r| < 1) \le q$, and e_i/r has a law in the class, which is closed under scaling; so $\mathrm{pr}(|V_i| < r) = \mathrm{pr}(|W/r + e_i/r| < 1) \le p_{\mathcal{C}}^{\zeta}(q)$ unit by unit, by the definition of $p_{\mathcal{C}}^{\zeta}(q)$ as a supremum, and N is stochastically smaller than $\mathrm{Bin}\{K, p_{\mathcal{C}}^{\zeta}(q)\}$. The transfer is applied at r itself rather than at $s < r$, because at threshold s the bound on the mean is $\zeta r/s > \zeta$; this is why the definition uses open intervals. With latent laws G_i that differ between units, the same holds when $G_i \in \mathcal{L}$, $Q_q(|W_i|) \ge r_0$ and $|E(W_i)| \le \zeta r_0$ for every i, with $r_0 = Q_q(|W_{\mathrm{new}}|)$, as in Corollary 1.

Part (ii). As $\mathcal{L} \subset \mathcal{B}$, Lemma S2 with Lemma S3 gives, for every $q' \in (q, 1)$, that noisy coverage above $\Psi_{\mathcal{C}}(q')$ forces $\mathrm{pr}(|W| \le 1) \ge q' > q$, hence $\mathrm{pr}(|W| < 1) > q$ for a law that is not a point mass; so feasible laws have noisy coverage at most $\Psi_{\mathcal{C}}(q')$, and letting $q' \downarrow q$, by continuity of $\Psi_{\mathcal{C}}$, $p_{\mathcal{C}}^{\zeta}(q) \le \Psi_{\mathcal{C}}(q)$. The law $W = 1 + s(a + \eta - X_1)$ of § S2.2, with $a = \log(1/q) < 1$, has mean $1 - s(1 - a - \eta) \in (-1, 1)$ for small s and η, so it is feasible for every $\zeta \ge 1$, and its noisy coverage tends to $E\{\min(1, q\mathrm{e}^{-\eta-\epsilon})\}$;

hence $p_{\mathcal{C}}^{\zeta}(q) = \Psi_{\mathcal{C}}(q)$ for $\zeta \geq 1$. For the strict lower bound, let W have density f proportional to $\mathrm{e}^{\beta w}$ on a segment $[a', b']$ with $a' < -1 < 1 < b'$, mean zero and $\mathrm{pr}(|W| < 1) = q$; for $\beta = 0$ the uniform law on $[-1/q, 1/q]$ qualifies, and by continuity so does a placement of the tilted law for small $\beta > 0$. As f is smooth near ± 1, with noise of variance v,

$$\mathrm{pr}(|W + e| < 1) - q = \tfrac{1}{2} v \{f'(1) - f'(-1)\} + O(v^2)$$

for Gaussian and for uniform noise, and $f'(1) - f'(-1) = \beta\{f(1) - f(-1)\} > 0$. So $p_{\mathcal{C}}^{0}(q) > q$, and $p_{\mathcal{C}}^{\zeta}(q)$ is nondecreasing in ζ. For $q > 1 - \mathrm{e}^{-1}$ only means in $(-1, 1)$ matter: if $E(W) = \mu \geq 1$, then $W + e$ is log-concave with mean μ for Gaussian or uniform noise, so $\mathrm{pr}(|W + e| < 1) \leq \mathrm{pr}(W + e < \mu) \leq 1 - \mathrm{e}^{-1} < q$ by fact 4 of § S1.

Part (iii). For a fixed noise law the noisy coverage is linear in the law of W; for symmetric unimodal noise $\epsilon = RU$, with U uniform on $[-1, 1]$ and $R \geq 0$ (§ S2.3), so the supremum over the class is over uniform laws, and for Gaussian noise over the scale. With a continuous noise law, $g(w) = \mathrm{pr}(|w + e| < 1)$ is bounded and continuous, and the constraints are $\int f_j \, d\mu \geq 0$ with $f_1 = q - 1_{(-1,1)}$, which is upper semicontinuous because the interval is open, and $f_2 = \zeta - w$, $f_3 = \zeta + w$. On a compact interval the localization theorem of Fradelizi and Guédon (2006) shows that the supremum of $\int g \, d\mu$ over the log-concave probabilities satisfying these constraints is attained at

a point mass or at a law with density $\mathrm{e}^{-\psi}$ on a segment, ψ the maximum of at most three affine functions. The reduction to compact support is as in Proposition S3, except that the mean must be kept: a feasible law with $\mathrm{pr}(|W| < 1) = q$ is first dilated slightly about its mean, which lowers its latent coverage strictly, then conditioned on $[-M, M]$ and translated back to its mean; for large M the result is feasible and its noisy coverage is arbitrarily close. The computation uses two pieces; a re-optimization over three pieces, which the theorem allows, changed no value by more than 6×10^{-11}.

Computation. A two-piece law on a segment is placed with its mean in $[0, \min(\zeta, 1))$, and scaled so that its latent coverage is exactly q; its distribution function, mean and $E\{(x - W)_+\}$ have closed forms, evaluated in log space. Noisy coverage under uniform noise is a combination of $E\{(x - W)_+\}$, and under Gaussian noise a Gauss–Legendre quadrature split at the kinks, accurate to 10^{-15} against 30-digit quadrature. A multistart Nelder–Mead search over the six parameters gives each value, and each maximizer is rechecked in 30-digit arithmetic. An adversarial search over 3700 log-concave laws outside the family, standard laws and random piecewise log-affine laws with 3 to 40 pieces, each optimized over its mean, scale and the noise scale, found none above the reported levels. The values are therefore

attained, hence necessary; that they are the suprema is numerical, without a certificate.

S3.5 A shifted new residual

Proposition S2. *Let W be log-concave with standard deviation ς, let $\mathrm{pr}(|W| \le t) \ge q'$ and $|\Delta| \le \rho\varsigma$. Then $\mathrm{pr}(|W + \Delta| \le t) \ge q' - \rho$. Conversely, for symmetric unimodal noise, $q_0 > \mathrm{e}^{-1}$ and $p = \Psi_{\mathcal{C}}(q_0)$, there are log-concave W_n, with standard deviations ς_n, and noise laws in the class such that, as $n \to \infty$, the noisy coverage $\mathrm{pr}(|W_n + e_n| \le t)$ tends to p, the latent coverage $\mathrm{pr}(|W_n| \le t)$ tends to $q_0 = \lambda_{\mathcal{C}}(p)$, and $\mathrm{pr}(|W_n + \rho\varsigma_n| \le t) \to q_0\mathrm{e}^{-\rho}$.*

Proof. A shift by $\Delta > 0$ moves the window $[-t, t]$ to $[-t - \Delta, t - \Delta]$, which loses at most the mass of $(t - \Delta, t]$, at most $\Delta \sup f$. For a log-concave density $\varsigma \sup f \le 1$, with equality for the exponential law (Bobkov and Chistyakov, 2015), so the loss is at most ρ. For the converse let $b = u^{-1}\log(1/q_0)$ and $W = t + b - X_u$, so that $\varsigma = 1/u$, $\mathrm{pr}(W \le t) = q_0$ and $\mathrm{pr}(|W| \le t) = q_0(1 - \mathrm{e}^{-2ut})$, and let the noise be uniform on $[-r/u, r/u]$ with r the maximizer of Theorem 1(i) at q_0. As $ut \to \infty$ the left end does not matter, the latent coverage tends to q_0 and, by scaling, the noisy coverage to $E\{\min(1, q_0\mathrm{e}^{-rU})\} = \Psi_{\mathcal{C}}(q_0) = p$, with U uniform on $[-1, 1]$;

and $\mathrm{pr}(W + \rho/u \le t) = \mathrm{pr}(X_u \ge b + \rho/u) = q_0 \mathrm{e}^{-\rho}$. □

The loss $q_0(1 - \mathrm{e}^{-\rho}) = q_0\rho + O(\rho^2)$ shows that the linear order of the bound cannot be improved. At $q = 0.9$, the latent coverage guaranteed by noisy coverage 0.9 under symmetric unimodal noise is thus between 0.8881 and 0.8892 for $\rho = 0.01$, and between 0.798 and 0.813 for $\rho = 0.1$. Measured relative to t rather than to ς, a shift has no nontrivial bound: point masses just inside the threshold satisfy every noisy constraint. A rule that picks the rank for latent level $q + \rho$ would protect against such shifts, but it is too coarse to be useful: under symmetric unimodal noise at $K = 1000$ it gives the ranks 918, 927, 945 and 962 for $\rho = 0$, 0.01, 0.03 and 0.05, and the last equals the rank without a shape assumption.

S4. What a known noise variance adds

When the noise is Gaussian with known variance D, the noisy threshold can be widened instead of its rank raised. This section treats that problem, which the main paper mentions only in its discussion. After scaling the threshold t to 1, write $x = D/t^2$, let $Z \sim N(0, 1)$ be independent of W and define

$$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{L},\ \mathrm{pr}(|W + x^{1/2}Z| \le 1) \ge p\}, \qquad \text{(S4.1)}$$

with $R_{p,q}(x) = -\infty$ if nothing is feasible. If calibration certifies noisy coverage p at threshold t, then $t\,R_{p,q}(D/t^2)$ is the smallest radius that guarantees latent coverage q for every log-concave latent law; $R^{\mathcal{B}}_{p,q} \geq R_{p,q}$ is defined with $\mathcal{B}$ in place of $\mathcal{L}$. A reduction to log-affine laws on segments makes $R_{p,q}$ computable (Proposition S3). The behaviour for small x is governed by the one-sided constant $c_q = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L},\ \mathrm{pr}(Y + Z \leq 0) \geq q\}$, attained along exponential tails and the same over $\mathcal{B}$; ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$.

Theorem S1. *For every $q \in (0,1)$, every $x > 0$ and every $W \in \mathcal{B}$ with $\mathrm{pr}(|W + x^{1/2}Z| \leq 1) \geq q$, $Q_q(|W|) \leq 1 + c_q x^{1/2}$. Moreover $R_{q,q}(x) = 1 + c_q x^{1/2} + o(x^{1/2})$ as $x \to 0$, and likewise for $R^{\mathcal{B}}_{q,q}$.*

The widening is therefore of the order of the noise standard deviation; at $q = 0.9$ it is below 1.16% of the radius for every $W \in \mathcal{B}$ (§ S4.4). A mean away from the boundary reduces the order to D: for a mean-zero latent law $Q_q(|W|) \leq (t^2 + D)^{1/2}$, via the heat equation for quantiles (Steinbrecher and Shaw, 2008), an order that centred log-affine laws attain, and the order $D^{1/2}$ returns only when the mean is within a bounded number of noise standard deviations of the boundary (Theorems S2 and S3); compare Jochmans and Weidner (2024) for a fixed smooth law. A noisy level of at least $\Psi_{\mathcal{N}}(q)$ removes the widening at every noise level (Proposition S4). Known vari-

ances also allow shorter intervals with heterogeneous units: a rule that is given the variances, assumes Gaussian noise and computes the radius for log-concave latent laws allowed by a single constraint on the average noisy kernel (HetLDC, § S4.7) was about 10% shorter than the rank of Theorem 3 at $K = 110$ under Gaussian noise, the only setting where it was run; its implementation computes an upper bound of the radius in double precision, which is not a certificate.

S4.1 Reduction to log-affine laws

Write $g_x(w) = \mathrm{pr}(|w + x^{1/2}Z| \le 1)$, so that the constraint in (S4.1) reads $E\{g_x(W)\} \ge p$, and let $\mathcal{F}$ be the family of point masses and of laws with density proportional to $\mathrm{e}^{\beta w}$ on a segment.

Proposition S3. *$R_{p,q}(x) = \sup\{Q_q(|W|) : W \in \mathcal{F},\ E\{g_x(W)\} \ge p\}$. The same holds for any continuous constraint kernel that vanishes at $\pm\infty$, in particular for the average kernel of § S4.7.*

Proof. Since $\mathcal{F} \subset \mathcal{L}$ it suffices to bound the supremum over $\mathcal{L}$. Let $W \in \mathcal{L}$ satisfy $E\{g_x(W)\} \ge p$ and let $s < Q_q(|W|)$, so that $\mathrm{pr}(|W| < s) \le q - \gamma$ for some $\gamma > 0$. Choose $M \ge s$ with $g_x(w) \le p$ for $|w| \ge M$, and let W_M be W conditioned on $|W| \le M$, which is log-concave. With $\varepsilon_M = \mathrm{pr}(|W| > M)$, the conditioning removes only mass on which the kernel

is at most p, so $E\{g_x(W_M)\} \geq (p - p\varepsilon_M)/(1 - \varepsilon_M) = p$ and $\mathrm{pr}(|W_M| < s) \leq (q-\gamma)/(1-\varepsilon_M) < q$ for M large. On $[-M, M]$ the constraint function $g_x - p$ is continuous, and $\mu \mapsto -\mu(-s, s)$ is linear and upper semicontinuous for weak convergence because $(-s, s)$ is open. By the localization theorem (Kannan, Lovász, and Simonovits, 1995; Fradelizi and Guédon, 2004), the supremum of this functional over the log-concave probability measures on $[-M, M]$ with $\int (g_x - p)\, d\mu \geq 0$ is attained at a point mass or at a law with log-affine density on a segment. Hence some member of $\mathcal{F}$ is feasible and has $\mathrm{pr}(|W| < s) < q$, that is $Q_q(|W|) \geq s$. Only continuity of the kernel and its decay at infinity were used. □

For $\beta > 0$ and $\sigma = x^{1/2}$, integration by parts gives

$$\int_a^b \mathrm{e}^{\beta w} \Phi\Big(\frac{c-w}{\sigma}\Big) dw = \frac{1}{\beta}\Big[\mathrm{e}^{\beta w} \Phi\Big(\frac{c-w}{\sigma}\Big)\Big]_a^b + \frac{\mathrm{e}^{\beta c + \beta^2\sigma^2/2}}{\beta}\Big\{\Phi\Big(\frac{b-c-\beta\sigma^2}{\sigma}\Big) - \Phi\Big(\frac{a-c-\beta\sigma^2}{\sigma}\Big)\Big\}, \tag{S4.2}$$

so the constraint over $\mathcal{F}$ is a combination of normal distribution functions. For fixed slope and length, the translations c of a member of $\mathcal{F}$ that satisfy the constraint form an interval, because $c \mapsto \mathrm{pr}(|W + c + e| \leq 1)$ is log-concave (Prékopa, 1973), and $c \mapsto Q_q(|W + c|)$ is quasi-convex, so it is maximized at an end of that interval.

S4.2 The one-sided problem

For $u > 0$ and the exponential tail $b - X_u$, independent of Z, let

$$m_u(b) = \mathrm{pr}(b - X_u + Z \le 0) = \Phi(-b) + \mathrm{e}^{-ub+u^2/2}\Phi(b-u),$$

which is continuous and strictly decreasing from 1 to 0 in b. Let $b_u(p)$ solve $m_u(b) = p$, and write $v_{p,q}(u) = b_u(p) - u^{-1}\log(1/q)$, $v_q = v_{q,q}$ and $k_q(u) = u^{-1} - b_u(q)$; the law $b_u(q) - X_u$ has q-quantile $v_q(u)$ and mean $-k_q(u)$.

Lemma S7. *Let* $c_{p,q} = \sup\{F_Y^{-1}(q) : Y \in \mathcal{L},\ \mathrm{pr}(Y + Z \le 0) \ge p\}$ *and* $c_q = c_{q,q}$. *Then* $c_{p,q} = \sup_{u>0} v_{p,q}(u)$, $c_q > 0$, *and* $q \mapsto c_q$ *is nonincreasing. The same holds with* $\mathcal{B}$, *or with the class of laws with* $\log F_Y$ *concave, in place of* $\mathcal{L}$. *Moreover* $c_{p,q} > 0$ *if and only if* $p < \Psi_{\mathcal{N}}(q)$.

Proof. Exponential tails are log-concave, which gives one inequality. Conversely, let $\log F_Y$ be concave with $\mathrm{pr}(Y + Z \le 0) \ge p$. Let $v = F_Y^{-1}(q)$. Concavity of $\log F_Y$ allows an atom only at the left end of the support. If Y has an atom there with mass at least q, a point mass included, then $Y \ge v$ almost surely, so $p \le \Phi(-v)$ and $v \le -\Phi^{-1}(p) = \lim_{u\to\infty} v_{p,q}(u)$. Otherwise $F_Y(v) = q$; let u be the right derivative of $\log F_Y$ at v, a supergradient; $u > 0$, since $u = 0$ would give $F_Y \le q < 1$. By fact 5 of § S1, $Y' = v + u^{-1}\log(1/q) - X_u$ lies stochastically below Y and has q-

quantile v, so $p \le \mathrm{pr}(Y+Z \le 0) \le m_u\{v + u^{-1}\log(1/q)\}$, and $v \le v_{p,q}(u)$ since m_u decreases. For monotonicity, let $q' > q$, let Y be feasible at (q', q') with $v = F_Y^{-1}(q')$, and put $\rho = (q'-q)/(1-q)$. Then Y conditioned on $Y \ge F_Y^{-1}(\rho)$ stays in the class: $\log(F-\rho) = h(\log F)$ with $h(z) = \log(\mathrm{e}^z - \rho)$ nondecreasing and concave, and the log survival function of the conditioned law is $\min\{0, \log(1-F) - \log(1-\rho)\}$, a minimum of concave functions. It satisfies the constraint at level $(q'-\rho)/(1-\rho) = q$, because the removed mass has noisy constraint value at most one, and has q-quantile v; point masses have value $-\Phi^{-1}(q') < -\Phi^{-1}(q) \le c_q$. For small u, $m_u(b) \ge \mathrm{e}^{-ub+u^2/2}\Phi(b-u) \to 1$ for each fixed b, so $b_u(q) \to \infty$; then $\mathrm{e}^{-ub+u^2/2}\Phi(b-u) \le q \le \Phi(-b) + \mathrm{e}^{-ub+u^2/2}$ at $b = b_u(q)$ gives $ub_u(q) \to a = \log(1/q)$, so $\Phi(-b)$ and $\Phi(u-b)$ are of order $\exp\{-a^2/(3u^2)\}$, and $-ub + u^2/2 = \log q + o(u^2)$. Hence

$$v_q(u) = u/2 + o(u), \qquad k_q(u) = (1-a)/u - u/2 + o(u) \qquad (u \to 0). \quad \text{(S4.3)}$$

Finally, at $b = u^{-1}\log(1/q)$, $m_u(b) = \Phi(-a/u) + q\mathrm{e}^{u^2/2}\Phi(a/u - u) = \Gamma(q,u)$, with $a = \log(1/q)$ and Γ as in (3.3). As m_u is strictly decreasing, $v_{p,q}(u) > 0$ if and only if $\Gamma(q,u) > p$, so $c_{p,q} > 0$ if and only if $\sup_u \Gamma(q,u) = \Psi_{\mathcal{N}}(q) > p$. In particular $c_q = c_{q,q} > 0$, since $\Psi_{\mathcal{N}}(q) > q$ (§ S3.1). □

The last statement links the two sections of the main paper: the expo-

nential tail with rate u and unit noise is the extremal law of Proposition 2 with noise scale u. Ball arithmetic gives $c_{0.8} = 0.04607$, $c_{0.9} = 0.01906$ and $c_{0.95} = 0.00841$ (§ S5).

S4.3 Proof of Theorem S1

Upper bound. Let $\alpha_R = \mathrm{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \mathrm{pr}(W + x^{1/2}Z < -1)$, so that $\alpha_L + \alpha_R \le 1 - q$. The law of $Y = (W-1)/x^{1/2}$ has $\log F_Y$ concave and $\mathrm{pr}(Y + Z \le 0) = 1 - \alpha_R$. By Lemma S7, $F_Y^{-1}(1-\alpha_R) \le c_{1-\alpha_R} \le c_q$, that is $\mathrm{pr}(W > 1 + c_q x^{1/2}) \le \alpha_R$. The same argument applied to $-W$, which uses concavity of $\log(1 - F_W)$, gives $\mathrm{pr}(W < -1 - c_q x^{1/2}) \le \alpha_L$, so $\mathrm{pr}(|W| \le 1 + c_q x^{1/2}) \ge q$.

Lower bound, for every $p \in (0,1)$. Fix $u > 0$, let $b_{\min} = b_u\{(1+p)/2\}$ and, for $p' \in [p, (1+p)/2]$, let $Y = b_u(p') - X_u$ and $W = 1 + x^{1/2}Y$, which is log-concave. With $\tau = 2x^{-1/2}$,

$$\mathrm{pr}(|W + x^{1/2}Z| \le 1) = p' - \mathrm{pr}(Y + Z < -\tau) \ge p' - \eta_x,$$

with $\eta_x = \mathrm{e}^{-u(\tau + b_{\min})/2} + \Phi\{-(\tau + b_{\min})/2\}$, by splitting the event $Y + Z < -\tau$ according to whether $Z < -(\tau + b_{\min})/2$. Take $p' = p + \eta_x$, which lies in $[p, (1+p)/2]$ for small x. Since $\mathrm{pr}(|W| \le s) \le F_W(s)$, $Q_q(|W|) \ge 1 + x^{1/2} v_{p+\eta_x, q}(u)$. As $\eta_x \to 0$ and b_u is continuous, $\liminf_{x \to 0}\{R_{p,q}(x) - 1\}/x^{1/2} \ge v_{p,q}(u)$ for every u, hence $\ge c_{p,q}$. With $p = q$ and the upper

bound this proves Theorem S1; the lower bound uses log-concave laws and so holds for $R^{\mathcal{B}}_{q,q}$ too.

S4.4 Feasibility and the size of the widening

The kernel g_x is symmetric and maximal at 0, so $E\{g_x(W)\} \le g_x(0) = 2\Phi(x^{-1/2}) - 1$ for every W, and a latent law is feasible at level q only if $x^{1/2} \le 1/\Phi^{-1}\{(1+q)/2\}$. With Theorem S1, the widening $Q_q(|W|) - 1 \le c_q x^{1/2}$ is at most $c_q/\Phi^{-1}\{(1+q)/2\}$, which is $0.01906/1.6449 < 1.16\%$ at $q = 0.9$, for every $W \in \mathcal{B}$ and every noise variance. Grid maximization over $\mathcal{F}$, which bounds $R_{q,q}$ from below, puts the largest widening at 0.37%, 0.063% and 0.012% for $q = 0.8$, 0.9 and 0.95, and the grid values fall below 1 beyond $x \approx 0.034$, 0.0062 and 0.0012 (numerical); grid values below 1 do not by themselves show that shrinking the threshold is safe.

S4.5 The noisy level that removes the widening

Proposition S4. *Let* $\mathrm{e}^{-1} < q < 1$. *If* $p \ge \Psi_{\mathcal{N}}(q)$, *then* $R^{\mathcal{B}}_{p,q}(x) \le 1$ *for every* $x > 0$. *If* $p < \Psi_{\mathcal{N}}(q)$, *then* $R_{p,q}(x) > 1$ *for all sufficiently small* $x > 0$.

Proof. If $p \ge \Psi_{\mathcal{N}}(q)$ and $W \in \mathcal{B}$ has $\mathrm{pr}(|W + x^{1/2}Z| \le 1) \ge p$, Lemma S3 for Gaussian noise gives $\mathrm{pr}(|W| \le 1) \ge \lambda_{\mathcal{N}}(p) \ge q$, so $Q_q(|W|) \le 1$. If $q \le p < \Psi_{\mathcal{N}}(q)$, then $c_{p,q} > 0$ by Lemma S7, and the lower bound of § S4.3

gives $R_{p,q}(x) \geq 1 + c_{p,q}x^{1/2} + o(x^{1/2}) > 1$ for small x; if $p < q$, $R_{p,q} \geq R_{q,q}$, which exceeds 1 for small x by Theorem S1. □

At $q = 0.9$ this gives the threshold $\Psi_{\mathcal{N}}(0.9) = 0.9008035$ stated above. In the same way noisy coverage p guarantees latent coverage $\lambda_{\mathcal{N}}(p)$ at every noise level, 0.89918 at $p = 0.9$, and no more for log-concave latent laws at small noise.

S4.6 The location of the mean

Let $R^{(b)}_{p,q}$ be defined as $R_{p,q}$ with the additional constraint $|E(W)| \leq b$, so that the mean is at least $1 - b$ from ± 1. A mean away from the boundary reduces the order of the widening from $x^{1/2}$ to x.

Theorem S2. *Let $W \in \mathcal{B}$ with $|E(W)| \leq M$, let $\sigma > 0$ and $t > M$, and suppose that* $\mathrm{pr}(|W + \sigma Z| \leq t) \geq q$. *Then*

$$Q_q(|W|) \leq M + \{(t - M)^2 + \sigma^2\}^{1/2} \leq t + \frac{\sigma^2}{2(t - M)}.$$

In particular $R^{(b)}_{p,q}(x) \leq 1 + x/\{2(1 - b)\}$ for all $p \geq q$ and $x > 0$, and a mean-zero latent law has $Q_q(|W|) \leq (t^2 + D)^{1/2}$.

Proof. Let $\mu = E(W)$. If W is a point mass, $Q_q(|W|) = |\mu| \leq M$. Otherwise, for $s > 0$ the law of $W + s^{1/2}Z$ is in $\mathcal{B}$, with mean μ, distribution function F_s and a positive smooth density f_s. For $r > \mu$ let $h = f_s(r)/F_s(r)$.

The tangent of $\log F_s$ at r gives $F_s(y) \le \min\{F_s(r)\mathrm{e}^{h(y-r)}, 1\}$, the distribution function of an exponential tail with mean $r - \{1 + \log F_s(r)\}/h$ that lies stochastically below $W + s^{1/2}Z$, so $\mu \ge r - \{1 + \log F_s(r)\}/h$ and $h(r-\mu) \le 1$. Laws in $\mathcal{B}$ satisfy $f_s' \le f_s^2/F_s$, so $(\log f_s)'(r) \le h \le 1/(r-\mu)$; applied to $-W$ this gives $(\log f_s)'(-r) \ge -1/(r+\mu)$. Hence, for $r > M$, $f_s'(r) \le f_s(r)/(r-M)$ and $-f_s'(-r) \le f_s(-r)/(r-M)$. Let $H_s(r) = \mathrm{pr}(|W + s^{1/2}Z| \le r)$. By the heat equation $\partial_s f_s = f_s''/2$,

$$\partial_s H_s(r) = \tfrac{1}{2}\{f_s'(r) - f_s'(-r)\} \le \frac{f_s(r) + f_s(-r)}{2(r-M)} = \frac{\partial_r H_s(r)}{2(r-M)}.$$

The quantile $r(s) = Q_q(|W + s^{1/2}Z|)$ solves $H_s\{r(s)\} = q$ and is continuously differentiable on $s > 0$, with $r'(s) \ge -1/[2\{r(s) - M\}]$ where $r(s) > M$ (Steinbrecher and Shaw, 2008). Hence $\{r(s)-M\}_+^2$ has derivative at least -1 almost everywhere, and $\{r(\varepsilon)-M\}_+^2 \le \{r(\sigma^2)-M\}_+^2 + \sigma^2 - \varepsilon$. As $\varepsilon \to 0$, $r(\varepsilon) \to Q_q(|W|)$, because a nondegenerate law in $\mathcal{B}$ has a positive density on the interior of its support, so the distribution function of $|W|$ is continuous and strictly increasing where it lies strictly between 0 and 1. Since $r(\sigma^2) \le t$, the first inequality follows, and the second from $(d^2 + \sigma^2)^{1/2} \le d + \sigma^2/(2d)$. □

The order x cannot be improved, even for mean-zero laws: if X has density proportional to $\mathrm{e}^{\beta x}$ on $[-\ell, 0]$, $W = X - E(X)$ and $r_0 = Q_q(|W|)$

with $\pm r_0$ interior to the support, then, after scaling W so that the noisy q-quantile of $|W+x^{1/2}Z|$ equals 1, $Q_q(|W|) = 1+\frac{1}{2}\beta r_0 \tanh(\beta r_0)x+o(x)$, because near $\pm r_0$ the density satisfies $f' = \beta f$ and $H_v(r) = H_0(r)+v\{f'(r)-f'(-r)\}/2 + o(v)$ uniformly near r_0. For every q the condition holds for small $\beta\ell$, so centring does not replace symmetry.

The order $x^{1/2}$ returns only when the mean is within a bounded number of noise standard deviations of the boundary. With $b_x = 1-\kappa x^{1/2}$, let

$$L_q(\kappa) = \sup\{F_Y^{-1}(q) : Y \in \mathcal{B},\ E(Y) \le -\kappa,\ \mathrm{pr}(Y+Z \le 0) \ge q\}.$$

Lemma S8. *Let* $\mathrm{e}^{-1} < q < 1$ *and* $\kappa \ge 0$. *Then* $L_q(\kappa) = \sup\{v_q(u) : u > 0,\ k_q(u) \ge \kappa\}$; L_q *is positive and nonincreasing; for* $q > 1/2$, $L_q(\kappa) = c_q$ *for* $\kappa \le \kappa_q^*$, *the largest value of* k_q *on the compact set of maximizers of* v_q; *and* $\kappa L_q(\kappa) \to \{1-\log(1/q)\}/2$ *as* $\kappa \to \infty$. *In double precision* $\kappa_q^* = 6.81$, 20.18 *and* 50.07 *at* $q = 0.8$, 0.9 *and* 0.95.

Proof. Let $a = \log(1/q)$ and $A_q = 1-a > 0$. For $q > 1/2$, v_q is continuous on $(0,\infty)$, tends to 0 as $u \to 0$ and to $-\Phi^{-1}(q) < 0$ as $u \to \infty$, and is positive somewhere, so its maximizers form a nonempty compact subset of $(0,\infty)$. If Y is nondegenerate with q-quantile v, the replacement of fact 5 of § S1, $Y' = v+a/u-X_u$, lies stochastically below Y, so $E(Y') \le E(Y) \le -\kappa$ and $\mathrm{pr}(Y'+Z \le 0) \ge q$, with the same q-quantile. For the tail $b-X_u$ the

constraints read $b \le b_u(q)$ and $b \le u^{-1} - \kappa$, so the largest quantile is $\min\{v_q(u), A_q/u - \kappa\}$. If $k_q(u) < \kappa$, this minimum is $A_q/u - \kappa$, and since k_q is continuous with $k_q(s) \to \infty$ as $s \to 0$, some $s < u$ has $k_q(s) = \kappa$ and value $A_q/s - \kappa$, larger; so only u with $k_q(u) \ge \kappa$ matter, and there the value is $v_q(u)$. A feasible point mass y_0 has $y_0 \le -\kappa \le 0 < L_q(\kappa)$, where positivity follows from (S4.3). Since $m_u(b) \ge \Phi(-b)$, $b_u(q) \ge -\Phi^{-1}(q)$, so k_q is bounded on $[u_0, \infty)$ for every $u_0 > 0$, and $k_q(u) \ge \kappa$ forces $u \le \{A_q + o(1)\}/\kappa$ as $\kappa \to \infty$; with $v_q(u) = \{1 + o(1)\}u/2$, by (S4.3), this gives $\limsup \kappa L_q(\kappa) \le A_q/2$. Conversely, for large κ there is u_κ with $k_q(u_\kappa) = \kappa$, by continuity, and $u_\kappa \to 0$; by (S4.3), $u_\kappa = \{A_q + o(1)\}/\kappa$ and $L_q(\kappa) \ge v_q(u_\kappa) = \{1 + o(1)\}u_\kappa/2$, so $\liminf \kappa L_q(\kappa) \ge A_q/2$. Monotonicity is clear, and $L_q(\kappa) = c_q$ while a maximizer of v_q is feasible. □

Theorem S3. *Let* $1 - \mathrm{e}^{-1} < q < 1$, $x > 0$ *and* $W \in \mathcal{B}$ *with* $\mathrm{pr}(|W + x^{1/2}Z| \le 1) \ge q$. *Then* $|E(W)| < 1$, *and if* $\Delta = Q_q(|W|) - 1 > 0$,

$$2\{1 - |E(W)|\}\Delta + \Delta^2 \le x. \tag{S4.4}$$

Moreover, for every $\kappa \ge 0$, $\lim_{x \to 0} x^{-1/2}\{R_{q,q}^{(b_x)}(x) - 1\} = L_q(\kappa)$, *and the same limit holds with* $\mathcal{L}$ *replaced by* $\mathcal{B}$ *in the definition of* $R^{(b)}$.

Proof. First part. Let $V = W + x^{1/2}Z$, with distribution function F in $\mathcal{B}$. By fact 4 of § S1, $F\{E(V)\} \ge \mathrm{e}^{-1}$ and $1 - F\{E(V)\} \ge \mathrm{e}^{-1}$. If $E(W) \ge 1$,

then $\mathrm{pr}(|V| \le 1) \le F\{E(W)\} \le 1 - \mathrm{e}^{-1} < q$, and similarly if $E(W) \le -1$; so $|E(W)| < 1$. Theorem S2 with $M = |E(W)|$ and $t = 1$ gives $\Delta \le (d^2 + x)^{1/2} - d$ with $d = 1 - |E(W)|$, which is (S4.4).

Lower bound in the limit. Take u with $k_q(u) \ge \kappa$ and the law $W = 1 + x^{1/2}\{b_u(q + \eta_x) - X_u\}$ of § S4.3. Its mean is $1 + x^{1/2}\{b_u(q + \eta_x) - u^{-1}\} \le 1 - \kappa x^{1/2}$, because b_u decreases in its argument, and it is positive for small x; hence the lim inf is at least $v_q(u)$ for every such u.

Upper bound in the limit. Suppose that $x_n \to 0$, $h_n = x_n^{1/2}$, and W_n is feasible for $R_{q,q}^{(b_{x_n})}(x_n)$ with $Q_q(|W_n|) - 1 \ge a_0 h_n$, where $L_q(\kappa) < a_0 < 1$. By (S4.4), $\kappa h_n \le 1 - |E(W_n)| \le \bar{\kappa} h_n$ with $\bar{\kappa} = (1 - a_0^2)/(2a_0)$. Replacing W_n by $-W_n$ if necessary, $E(W_n) \ge 0$; let $Y_n = (W_n - 1)/h_n$, so that $-\bar{\kappa} \le E(Y_n) \le -\kappa$. The law of $T_n = Y_n + Z$ is in $\mathcal{B}$ with a smooth positive density of maximum M_n and distribution function F_n, and $F_n(0) \ge q$ while $F_n\{E(T_n)\} \le 1 - \mathrm{e}^{-1}$, so $q - 1 + \mathrm{e}^{-1} \le F_n(0) - F_n\{E(T_n)\} \le M_n \bar{\kappa}$. For a law in $\mathcal{B}$ with smooth density g, maximum M and median m, the monotone hazards give $g(m) \ge M/2$ and $\mathrm{pr}(|T - m| > y) \le \mathrm{e}^{-2g(m)y}$, hence $\mathrm{var}(T) \le 2/M^2$. Thus the $\mathrm{var}(Y_n)$ are bounded, the Y_n are tight and uniformly integrable, and along a subsequence $Y_n \to Y$ in distribution with $Y \in \mathcal{B}$, possibly a point mass, $E(Y) \le -\kappa$ and $\mathrm{pr}(Y + Z \le 0) \ge q$. On $\{W_n \ge 0\}$, whose probability tends to one, $(|W_n| - 1)/h_n = Y_n$, so

$a_0 \le \{Q_q(|W_n|) - 1\}/h_n \to F_Y^{-1}(q) \le L_q(\kappa)$ by Lemma S8, where the quantile converges because a law in $\mathcal{B}$ has a unique q-quantile or is a point mass. This contradicts $a_0 > L_q(\kappa)$; if $L_q(\kappa) \ge 1$ there is nothing to prove, because (S4.4) gives $\Delta < x^{1/2}$. □

At $q = 0.9$, L_q equals c_q up to $\kappa \approx 20$ and then decays like $1/\kappa$: $L_{0.9}(25) = 0.017393$ and $L_{0.9}(50) = 0.0089448$ (double precision). A mean near the boundary is necessary for a widening of order $x^{1/2}$, not sufficient: a Gaussian law with mean $1.29x^{1/2}$ from the boundary after scaling needs no widening at all.

S4.7 Rules that use known variances

The rules of this subsection use the noise variances D_i and assume Gaussian noise; the Markov rule is in Table 6, and HetLDC and the rule with a known lower bound on the variances are in the full tables of § S6.4.

An order-statistic level. Let $T = |V|_{(k)}$, let $\bar{F}$ be the average distribution function of the $|V_i|$, assumed continuous, and let p_k be the δ-quantile of the Beta$(k, K + 1 - k)$ distribution.

Proposition S5. *If* $k \ge Kp_k + 1$, *then* $\mathrm{pr}\{\bar{F}(T) \ge p_k\} \ge 1 - \delta$.

Proof. $\bar{F}(T) < p_k$ exactly when $N = \#\{i : |V_i| < \bar{F}^{-1}(p_k)\} \ge k$. N is a sum of independent indicators whose success probabilities average p_k, and

$k-1 \geq Kp_k$, so by Hoeffding (1956) $\mathrm{pr}(N \leq k-1)$ is smallest when all success probabilities are equal. Hence $\mathrm{pr}(N \geq k) \leq \mathrm{pr}\{\mathrm{Bin}(K, p_k) \geq k\} = \mathrm{pr}\{\mathrm{Beta}(k, K+1-k) \leq p_k\} = \delta$. □

HetLDC: the radius for the average kernel. Let $\bar{g}(w) = K^{-1} \sum_i \mathrm{pr}(|w + e_i| \leq T)$ be the average noisy kernel at T. On the event of Proposition S5, $\int \bar{g}\, dG \geq p_k$. Let R^{mix} be defined by (S4.1), in units in which the threshold is 1, with the constraint $\int \bar{g}(Tz)\, dG_T(z) \geq p_k - \omega$ on the law G_T of W/T, where ω bounds the sup-norm error of compressing the D_i to 32 support points, about 2×10^{-4}. By Proposition S3, which holds for this kernel, the radius TR^{mix} covers the latent target with probability at least q, given the data, on an event of probability at least $1-\delta$, for every log-concave G. This guarantee is for the exact R^{mix}; the implementation computes an upper bound on it by a branch and bound over $\mathcal{F}$ whose closed forms (S4.2) are evaluated in double precision with a margin of 10^{-9}, which is not a certificate. Replacing the average kernel by the kernel of a single variance is not safe: with the mean variance the radius is 1.4% too short at mean $x = 0.2$ with lognormal spread 1 (numerical), and the smallest variance also fails.

The rule with a known lower bound on the variances uses a two-sided version of the constant of Lemma S7. With $c_{1,q'} = -\infty$, since no law has

noisy coverage one at a single end, and $c_{p',1} = \infty$, let

$$C_{p,q} = \sup_{\alpha_L+\alpha_R\le 1-p} \inf_{\beta_L+\beta_R=1-q} \max(c_{1-\alpha_L,1-\beta_L},\, c_{1-\alpha_R,1-\beta_R}),$$

over nonnegative α's and β's. Taking $\alpha_L = 0$ shows $C_{p,q} \ge c_{p,q}$.

Lemma S9. *For every $x > 0$ and every $W \in \mathcal{B}$ with $\mathrm{pr}(|W + x^{1/2}Z| \le 1) \ge p$, $Q_q(|W|) \le \max\{0, 1 + C_{p,q}x^{1/2}\}$.*

Proof. Let $\alpha_R = \mathrm{pr}(W + x^{1/2}Z > 1)$ and $\alpha_L = \mathrm{pr}(W + x^{1/2}Z < -1)$, so $\alpha_L + \alpha_R \le 1 - p$, and for $\varepsilon > 0$ take $\beta_L + \beta_R = 1 - q$ with both constants at most $C_{p,q} + \varepsilon$. The law of $Y = (W - 1)/x^{1/2}$ has $\log F_Y$ concave and $\mathrm{pr}(Y + Z \le 0) = 1 - \alpha_R$, so by Lemma S7 $F_Y^{-1}(1 - \beta_R) \le C_{p,q} + \varepsilon$ and $\mathrm{pr}\{W > 1 + (C_{p,q} + \varepsilon)x^{1/2}\} \le \beta_R$. The same argument for $-W$ gives $\mathrm{pr}\{W < -1 - (C_{p,q} + \varepsilon)x^{1/2}\} \le \beta_L$, so $\mathrm{pr}\{|W| \le 1 + (C_{p,q} + \varepsilon)x^{1/2}\} \ge q$; let $\varepsilon \to 0$. □

Corollary S1. *Let $G \in \mathcal{B}$, let the noise be Gaussian and $k \ge Kp_k + 1$. If $D_i \ge D_{\min}$ for all i and $C_{p',q} < 0$ for some p' with $q \le p' \le p_k$, then $s = \max\{0, T + C_{p',q}D_{\min}^{1/2}\}$ satisfies $\mathrm{pr}_{\mathcal{D}}\{\mathrm{pr}(|W_{\mathrm{new}}| \le s \mid \mathcal{D}) \ge q\} \ge 1 - \delta$.*

Proof. On the event of Proposition S5 some i has $\mathrm{pr}(|W + e_i| \le T) \ge p_k \ge p'$, and Lemma S9 at threshold T and $x = D_i/T^2$ gives $Q_q(|W|) \le \max\{0, T + C_{p',q}D_i^{1/2}\} \le \max\{0, T + C_{p',q}D_{\min}^{1/2}\}$, as $C_{p',q} < 0$. □

At $q = 0.9$, $K = 110$ and $\delta = 0.05$, $p_k = 0.906806$ and ball arithmetic gives $C_{0.9068,0.9} \leq -0.114$, hence the known-lower-bound rule $s = T - 0.114 D_{\min}^{1/2}$. At $K = 1000$ the order-statistic level of the rank used, 917, lies below the smallest level with a certified negative constant, 0.9036, so the rule returns T itself, which is why its width equals that of the Gaussian rank in Table 6.

Proposition S6. *Let $\bar{g}$ be a kernel that is symmetric, continuous, strictly decreasing in $|w|$ and tends to 0 as $|w| \to \infty$, with $m_0 = \bar{g}(0)$, and suppose that $qm_0 < p < m_0$. Over all laws G on the real line with $\int \bar{g}\, dG \geq p$, the supremum of $Q_q(|W|)$ is the radius r that solves $\bar{g}(r) = (p - qm_0)/(1 - q)$.*

Proof. Let $\tau = (p - qm_0)/(1 - q) \in (0, m_0)$. Since $m_0 - \bar{g}(W) \geq 0$, Markov's inequality gives $\mathrm{pr}\{\bar{g}(W) < \tau\} \leq (m_0 - p)/(m_0 - \tau) = 1 - q$, and $\bar{g}(W) < \tau$ exactly when $|W| > r$, so $Q_q(|W|) \leq r$. Conversely, mass $q - \epsilon$ at 0 and $1 - q + \epsilon$ at the point $w_\epsilon > 0$ with $\bar{g}(w_\epsilon) = \{p - (q - \epsilon)m_0\}/(1 - q + \epsilon)$ meets the constraint with equality and has $Q_q(|W|) = w_\epsilon \to r$. □

The Markov rule of § 5.3 is the radius of Proposition S6 with the average kernel at the level p_k of a rank fixed in advance, the rank for level 0.925; it is the smallest radius that this single constraint allows without a shape assumption. With 1 in place of $\bar{g}(0)$ it contains the bound of Guille-Escuret

and Ndiaye (2024).

S5. Certificates in ball arithmetic

Three levels of numerical support are kept apart. Certificates in Arb ball arithmetic (Johansson, 2017), through `python-flint`, are proofs: every deciding inequality is evaluated on balls, floating point only proposes points, balls are converted to doubles with outward rounding, and sums of bounds are formed as balls. Upper bounds evaluated in double precision with a margin are labelled numerical. Grid and search values are used only as lower bounds or as proposals.

Gaussian levels (`latentcov.gaussian`). By Lemma S4, A is decreasing, so bisection on balls brackets b^* by doubles $b_{\mathrm{lo}} < b^* < b_{\mathrm{hi}}$ with $A(b_{\mathrm{lo}}) > \log(1/q) > A(b_{\mathrm{hi}})$ proved. The lower end of the enclosure of $\Psi_{\mathcal{N}}(q)$ is $G\{q, m(b_{\mathrm{lo}})\}$, a value of the supremand; the upper end is $\Phi(-c)+\phi(c)/m(b)$ at $b = b_{\mathrm{hi}}$, which equals $\Psi_{\mathcal{N}}(q_{\mathrm{hi}})$ with $q_{\mathrm{hi}} = \mathrm{e}^{-A(b_{\mathrm{hi}})} > q$, and $\Psi_{\mathcal{N}}$ increases. The enclosures have width below 3×10^{-16}; at $q = 0.9$, $\Psi_{\mathcal{N}}(0.9) = 0.900803469446498$.

Symmetric unimodal levels (`latentcov.noise_classes.psi_su_enclosure`). The root r of d in § S2.3 is bracketed in the same way, using that d decreases, and g is evaluated on balls at both ends of the

bracket; the split certificate `split_certificate_su` checks the two-sided level independently.

Mean-zero log-concave levels (`lc_certificate_parallel`, `split_certificate_lc`). The branch and bound of § S2.4 evaluates the cut bound on balls, with box ends entering as exact doubles and derived ends such as $1/|k|$ rounded outward; near $k = 0$, Z_k and $\int_0^1 v\mathrm{e}^{kv}\,dv$ are evaluated by their series with a remainder bound. The thresholds for the ends of the scale parameter are found by enlarging the parameter until the analytic bound holds on a ball.

One-sided constants (`latentcov.interval.c_enclosure`). The function $m_u(b)$ decreases in b, and in u because X_u is stochastically decreasing in its rate; hence $b_u(p)$ is nonincreasing in u and, on $[u_0, u_1]$, $v_{p,q}(u) \le b_{u_0}(p) - u_1^{-1}\log(1/q)$. A ball with $m_u(b) < p$ proves $b_u(p) < b$. For $u \ge u_{\max}$ the value is at most $b_{u_{\max}}(p)$; for small u, with $b(u) = \{\log(1/p) + u^2/2 - \log(1-u^3)\}/u$, $\mathrm{e}^{-ub+u^2/2} = p(1-u^3)$ and the Mills ratio gives $\Phi(-b) \le pu^3$ whenever $\phi\{\log(1/p)/u\} \le pu^2\log(1/p)$, which holds on $(0, u_{\min}]$ once it holds at $u_{\min} \le \log(1/p)/2^{1/2}$; so $b_u(p) \le b(u)$ there. The certificates give $c_{0.8} \in [0.0460730, 0.0460731]$, $c_{0.9} \in [0.0190618, 0.0190619]$ and $c_{0.95} \in [0.0084075, 0.0084077]$. For the two-sided constant of Lemma S9: by Lemma S7, $c_{p',q'} \le \gamma$ if and only if $1 - q' \ge \varphi_\gamma(1 - p')$, with $\varphi_\gamma(\alpha) =$

$\max[0, \sup_u 1-\exp\{-u(b_u(1-\alpha)-\gamma)\}]$, the largest latent failure beyond γ at one end with noisy failure α there. Hence $C_{p,q} \le \gamma$ if $\varphi_\gamma(\alpha_L)+\varphi_\gamma(\alpha_R) \le 1-q$ whenever $\alpha_L + \alpha_R = 1 - p$; since φ_γ is nondecreasing, it suffices to check this at the ends of an adaptive partition of α_L, with φ_γ bounded above by the monotone branch and bound in u. This gives $C_{0.9036,0.9} \le -0.057$ and $C_{0.9068,0.9} \le -0.114$.

The bimodal law (`latentcov.interval.bimodal_blc_certificate`). The two inequalities of Lemma S1 are checked on 128 boxes covering $[a, m_2]$ and $[m_1, b]$, with a and b enclosed by bisection on balls and rounded outward.

Double precision only. The transition values κ_q^* and $L_q(\kappa)$, the grid maxima of § S4.4 and the upper bounds of R^{mix} used by HetLDC are computed in double precision and are not certified.

S6. Numerical studies: design and full results

Throughout, $\delta = 0.05$, latent reliability is the probability over calibration samples that the latent coverage given the data is at least q, and latent coverage given the data is computed from the exact latent distribution function. Ranks: the marginal rank is $\lceil q(K+1) \rceil$; the usual rank is the smallest k with $\mathrm{pr}\{\mathrm{Bin}(K, q) \le k-1\} \ge 1-\delta$; the class ranks use the levels

of Table 2.

S6.1 Exact reliability at the extremal law

With the threshold scaled to 1 and Gaussian noise $N(0, s^2)$, $s = 10^{-3}$, the latent law is the exponential tail $W = 1+(s/u)\{\log(1/q)-X_1\}$, with $u = s^*$ the maximizer of Theorem 2. Then $\mathrm{pr}(W \le 1) = q$, the latent q-quantile of $|W|$ is $r_q = 1$ to machine precision because the left end is $2/s$ noise standard deviations away, and the noisy coverage at r_q is $H(r_q) = \Psi_{\mathcal{N}}(q)$ to 10^{-14}. As in the proof of Theorem 3, the latent reliability of rank k is exactly $\mathrm{pr}\{\mathrm{Bin}(K, H(r_q)) \le k-1\}$. Table S4 gives the exact values behind Figure 1; the curves of Figure 1 use 95 values of K between 30 and 10^7.

S6.2 Mixed noise laws

The latent law is the extremal exponential tail of § S6.1 in the small-noise limit, where the left end does not matter. In units of the tail scale, a unit whose noise is $c\epsilon$, with ϵ of unit variance, has noisy coverage $h(\epsilon, c) = E\{\min(1, q\mathrm{e}^{-c\epsilon})\}$ at the latent q-quantile, by Proposition 2. Each unit draws its noise law from a mixture within one class and its scale $c = c_0 L$, with $L = 1$ (equal scales) or L lognormal with log-scale 0.7 and mean 1 (heterogeneous scales), independently. The $|V_i|$ are then independent and identically

Table S4: Exact latent reliability at the extremal law for Gaussian noise: rank and reliability

		marginal		usual		Theorem 3, Gaussian		symm. unimodal	
q	K	k	rel.	k	rel.	k	rel.	k	rel.
0.8	110	89	0.492	96	0.959	96	0.959	96	0.959
	10^3	801	0.375	822	0.915	826	0.955	829	0.974
	10^4	8001	0.136	8067	0.713	8110	0.951	8143	0.994
	10^5	80001	0.000	80209	0.032	80648	0.950	80978	1.000
	10^6	800001	0.000	800659	0.000	805064	0.950	808374	1.000
	10^7	8000001	0.000	8002081	0.000	8046170	0.950	8079296	1.000
0.9	110	100	0.535	105	0.967	105	0.967	105	0.967
	10^3	901	0.482	916	0.943	917	0.954	918	0.964
	10^4	9001	0.399	9050	0.918	9058	0.952	9068	0.977
	10^5	90001	0.199	90157	0.790	90237	0.951	90334	0.996
	10^6	900001	0.004	900494	0.150	901296	0.950	902278	1.000
	10^7	9000001	0.000	9001561	0.000	9009590	0.950	9019422	1.000
0.95	110	106	0.646	109	0.976	109	0.976	109	0.976
	10^3	951	0.511	962	0.955	962	0.955	962	0.955
	10^4	9501	0.477	9537	0.947	9538	0.952	9541	0.964
	10^5	95001	0.411	95114	0.922	95130	0.951	95157	0.980
	10^6	950001	0.233	950359	0.820	950518	0.950	950789	0.998
	10^7	9500001	0.010	9501134	0.254	9502722	0.950	9505441	1.000

The symmetric unimodal rank is valid here, since Gaussian noise is symmetric unimodal, and conservative. Scanning every integer K, the usual rank's reliability first falls below 0.9 at $K = 1122$, 18,555 and 248,781 and below 0.5 at $K = 22{,}177$, 376,330 and 5,080,045 for $q = 0.8$, 0.9 and 0.95.

distributed, and the latent reliability of rank k is $\mathrm{pr}\{\mathrm{Bin}(K, H) \le k-1\}$ exactly, with $H = E\,h(\epsilon, c_0 L)$ computed by quadrature and Gauss–Hermite integration over L. The common scale c_0 is chosen adversarially, to maximize H. The mixtures are: within the symmetric unimodal class, uniform noise with weight 1, 1/2 or 1/4 and the rest split equally between Gaussian, Laplace and t_3 noise, at $q = 0.8$ and 0.9; within the mean-zero log-concave class, the centred exponential law $X_1 - 1$ with weight 1, 1/2 or 1/4 and the rest split between Gaussian, uniform, Laplace and the reflected law $1 - X_1$, and the two signs with weight 1/2 each, at $q = 0.9$. That gives 20 cases.

Table S5: Mixed noise laws at the extremal latent law: worst noisy coverage H at the latent q-quantile, its excess over q, and exact latent reliability of the usual rank and of the class rank of Theorem 3

						$K = 10^3$		$K = 10^4$		$K = 10^5$	
class	q	mixture	heterog. scales	H	$10^3(H-q)$	usual	class	usual	class	usual	class
SU	0.8	uniform	no	0.807725	7.725	0.866	0.954	0.392	0.952	0.000	0.950
SU	0.8	uniform	yes	0.801887	1.887	0.942	0.984	0.884	0.999	0.562	1.000
SU	0.8	uniform 1/2, rest	no	0.805158	5.158	0.905	0.970	0.646	0.989	0.007	1.000
SU	0.8	uniform 1/2, rest	yes	0.801407	1.407	0.946	0.985	0.906	0.999	0.704	1.000
SU	0.8	uniform 1/4, rest	no	0.803924	3.924	0.920	0.976	0.753	0.996	0.072	1.000
SU	0.8	uniform 1/4, rest	yes	0.801179	1.179	0.948	0.986	0.915	1.000	0.764	1.000
SU	0.9	uniform	no	0.901787	1.787	0.930	0.955	0.856	0.953	0.406	0.950
SU	0.9	uniform	yes	0.900224	0.224	0.949	0.968	0.943	0.986	0.922	1.000
SU	0.9	uniform 1/2, rest	no	0.900828	0.828	0.942	0.964	0.917	0.977	0.782	0.996
SU	0.9	uniform 1/2, rest	yes	0.900137	0.137	0.950	0.969	0.947	0.987	0.934	1.000
SU	0.9	uniform 1/4, rest	no	0.900514	0.514	0.946	0.966	0.932	0.982	0.867	0.999
SU	0.9	uniform 1/4, rest	yes	0.900106	0.106	0.950	0.969	0.948	0.987	0.938	1.000
LC	0.9	centred exp.	no	0.905176	5.176	0.868	0.964	0.467	0.974	0.000	0.995
LC	0.9	centred exp.	yes	0.901127	1.127	0.939	0.987	0.900	1.000	0.678	1.000
LC	0.9	centred exp. 1/2, rest	no	0.900746	0.746	0.943	0.988	0.921	1.000	0.807	1.000
LC	0.9	centred exp. 1/2, rest	yes	0.900198	0.198	0.949	0.990	0.944	1.000	0.926	1.000
LC	0.9	centred exp. 1/4, rest	no	0.900458	0.458	0.947	0.989	0.934	1.000	0.879	1.000
LC	0.9	centred exp. 1/4, rest	yes	0.900113	0.113	0.950	0.990	0.947	1.000	0.937	1.000
LC	0.9	both signs, 1/2 each	no	0.900236	0.236	0.949	0.989	0.943	1.000	0.920	1.000
LC	0.9	both signs, 1/2 each	yes	0.900065	0.065	0.951	0.990	0.949	1.000	0.943	1.000

SU, symmetric unimodal class, level 0.8077245 ($q = 0.8$) or 0.9017873 ($q = 0.9$); LC, mean-zero log-concave class, sufficient level 0.906. "Rest": the remaining weight split equally between the other laws of the class listed in the text.

In every case H never exceeds the class level, as Theorems 1 and 3 imply;

the pure uniform case reproduces $\Psi_{\mathcal{C}}(q)$ to 10^{-16} and the pure centred exponential case gives 0.9051755. The class rank has reliability at least 0.9501 in all 20 cases and at all K up to 10^6, with the minimum, 0.950198, at pure uniform noise with equal scales. Heterogeneous scales dilute the excess $H - q$ because one common c_0 cannot put every unit at its worst scale: from 1.8×10^{-3} to 2.2×10^{-4} for uniform noise at $q = 0.9$. They delay the failure of the usual rank but do not remove it.

S6.3 Rules with the same information

The six latent laws have mean zero and variance one: normal; Laplace; centred gamma with shape 2; the truncated exponential law with density proportional to e^{4y} on $[0, 1]$, standardized; the bimodal law of § S1; and t_3, outside $\mathcal{B}$ because its polynomial tails make $\log(1 - F)$ convex far out. The noise variances are $D_i = 0.577 L_i / E(L_i)$, L_i lognormal with log-scale 0.7, drawn afresh for each data set, and each unit's noise law is scaled to variance D_i: uniform on $[-(3D_i)^{1/2}, (3D_i)^{1/2}]$, Laplace with scale $(D_i/2)^{1/2}$, t_3 multiplied by $(D_i/3)^{1/2}$, and $D_i^{1/2}(X_1 - 1)$ for the centred exponential law. The eight noise designs are the five single laws and laws drawn per unit uniformly from {Gaussian, uniform, Laplace, t_3}, from {Gaussian, uniform, Laplace, $X_1 - 1$, $1 - X_1$} and from all six. A rule covers a setting when

every law of the design is in its noise class; the bi-log-concave rules also need the latent law in $\mathcal{B}$. There are 2000 data sets per setting. The extra width of a rule is $100(\bar{w}/\bar{w}_0 - 1)$, with $\bar{w}$ its mean width and $\bar{w}_0$ that of the usual rank, and its saving over the rule without a shape assumption of the same noise class is computed in the same way. Table S6 gives the results for both K and all three q.

S6.4 Comparison with existing methods

Design. The latent laws are those of § S6.3, the variances are drawn as there, $q = 0.9$, and there are 300 data sets at $K = 110$ and 200 at $K = 1000$. The noise is Gaussian, Laplace, or drawn per unit from {Gaussian, uniform, Laplace, t_3}, with the same variances. Every rule outputs a latent interval whose coverage given the data is exact. A new unit has no variance of its own, so the rules that need one draw it from the law that generated the D_i, with 400 draws per data set, and coverage and width are averaged over the draws.

Rules. (i) The usual rank and the ranks of Theorem 3 for Gaussian and symmetric unimodal noise, and the rank at level $(1 + q)/2$ without a shape assumption; none uses the variances. (ii) LatentCP (Zheng, Zhou, and Zhu, 2026), single level: the noisy conformal threshold t_γ at level $1 - \gamma$

Table S6: Rules with the same information: rank, smallest reliability over the settings a rule covers, width above the usual rank and below the rule without a shape assumption of the same noise class, in percent (ranges; medians in parentheses)

q	K	rule	k	reliability	above usual	below no shape
0.8	110	usual	96	1.000	–	–
		Gaussian	96	1.000	0.0	22–29 (28)
		symm. unimodal	96	1.000	0.0	21–32 (28)
		no shape, median zero	105	1.000	26.3–51.4	–
		no shape, log-concave	107	1.000	37.1–77.6	–
	1000	usual	822	1.000	–	–
		Gaussian	826	1.000	0.9–1.2	20–26 (24)
		symm. unimodal	829	1.000	1.5–2.2	19–28 (24)
		no shape, median zero	916	1.000	25.3–42.4	–
		no shape, log-concave	941	1.000	35.7–63.1	–
0.9	110	usual	105	1.000	–	–
		Gaussian	105	1.000	0.0	21–28 (25)
		symm. unimodal	105	1.000	0.0	18–33 (27)
		mean-zero log-concave	105	1.000	0.0	27–48 (39)
		no shape, median zero	109	1.000	22.5–62.9	–
		no shape, log-concave	110	1.000	36.9–121.8	–
	1000	usual	916	1.000	–	–
		Gaussian	917	1.000	0.3–0.4	16–21 (21)
		symm. unimodal	918	1.000	0.6–1.0	15–24 (20)
		mean-zero log-concave	922	1.000	1.8–2.9	19–33 (26)
		no shape, median zero	962	1.000	17.7–35.9	–
		no shape, log-concave	974	1.000	25.4–54.3	–
0.95	110	usual	109	0.999	–	–
		Gaussian	109	1.000	0.0	–
		symm. unimodal	109	0.999	0.0	–
		no shape, median zero	–	–	–	–
		no shape, log-concave	–	–	–	–
	1000	usual	962	1.000	–	–
		Gaussian	962	1.000	0.0	14–20 (17)
		symm. unimodal	962	1.000	0.0	12–23 (18)
		no shape, median zero	984	1.000	14.2–34.8	–
		no shape, log-concave	989	1.000	19.7–49.0	–

The usual rank's reliability is over all 48 settings, which it does not cover. At $K = 110$ and $q = 0.95$ the rules without a shape assumption have no valid rank. The mean-zero log-concave level is certified only at $q = 0.9$.

with $\gamma = \alpha/2$, $\alpha = 0.1$, inverted through the Gaussian forward model with the new unit's variance d to $\{w : \{1 - p_\gamma(w, d)\}/\gamma \leq 1/\alpha\}$, with $p_\gamma(w, d) = \mathrm{pr}(|w + d^{1/2}Z| \leq t_\gamma)$; tuned versions with one or two levels, the levels and weights chosen on a random half of the data by mean width and calibrated on the other half (for $K = 1000$ the grid is thinned to 12 levels and the width criterion uses 100 tuning variances, for computing time); and a version with the threshold at a high-probability rank, our adaptation. (iii) The Markov rule of Proposition S6 with the known variances. (iv) A Fay–Herriot interval: an intercept-only Fay–Herriot model fitted by restricted maximum likelihood with the D_i known, and a tolerance multiplier from a parametric bootstrap with 300 replicates, the bootstrap $(1 - \delta)$-quantile of the multiplier each refit needs to cover $N(\hat{\mu}, \hat{A})$ with probability 0.9, adapted from Chatterjee, Lahiri, and Li (2008). (v) HetLDC, the radius of § S4.7, run at $K = 110$ under Gaussian noise only, because its branch and bound takes about 2.5 minutes of one core per data set. (vi) The known-lower-bound rule of Corollary S1. (vii) The deconvolution threshold of Cohen, Goldberger, and Tirer (2025), Algorithm 1, in our implementation: a histogram of the noisy scores with bin width h, a deconvolution by nonnegative least squares with the average Gaussian kernel and ridge penalty 0.01, and the threshold lowered in steps of 0.01 from the noisy con-

formal threshold while the estimated latent coverage stays at least 0.9; with $h = 0.01$ empty bins are masked, as in their algorithm, and with $h = 0.2$ they are fitted as zeros. The tuning grids of LatentCP and the bin widths and scan step of the deconvolution threshold are our choices; the original papers do not fix them for this setting.

Table S7: Comparison with existing methods at $K = 110$: smallest mean latent coverage, smallest reliability and range of width relative to the usual rank (%), over the five bi-log-concave latent laws, by noise scenario

	Gaussian noise			Laplace noise			mixed symmetric unimodal		
rule	mean cov.	rel.	width	mean cov.	rel.	width	mean cov.	rel.	width
Theorem 3, Gaussian	0.969	1.000	0.0	0.971	1.000	0.0	0.968	1.000	0.0
Theorem 3, symm. unimodal	0.969	1.000	0.0	0.971	1.000	0.0	0.968	1.000	0.0
no shape, median zero	0.990	1.000	26.5–38.8	0.991	1.000	31.9–40.8	0.990	1.000	30.7–41.9
LatentCP	0.974	1.000	4.1–5.8	0.977	1.000	5.0–6.8	0.973	1.000	4.4–6.6
LatentCP, tuned	0.973	0.987	4.1–12.9	0.975	0.983	6.1–12.4	0.972	0.983	7.3–19.3
LatentCP, two levels	0.972	0.980	5.3–13.3	0.975	0.977	7.1–13.6	0.973	0.977	10.7–14.6
LatentCP, PAC rank	0.990	1.000	26.5–38.8	0.991	1.000	31.9–40.8	0.990	1.000	30.7–41.9
Markov rule	0.985	1.000	19.1–23.2	0.986	1.000	20.8–24.8	0.985	1.000	19.1–24.3
Fay–Herriot	0.936	0.937	-23.0 to -17.9	0.934	0.927	-24.0 to -17.3	0.932	0.900	-23.4 to -15.2
HetLDC	0.957	0.983	-10.9 to -9.5	–	–	–	–	–	–
known lower bound	0.968	1.000	-1.3 to -1.2	0.970	1.000	-1.3 to -1.2	0.966	1.000	-1.3
deconvolution, bin 0.01	0.279	0.000	-84.5 to -81.9	0.280	0.000	-84.7 to -81.8	0.278	0.000	-84.5 to -81.6
deconvolution, bin 0.2	0.894	0.537	-36.8 to -29.6	0.880	0.453	-40.9 to -32.2	0.889	0.517	-38.9 to -31.3

Table S8: As Table S7, at $K = 1000$

	Gaussian noise			Laplace noise			mixed symmetric unimodal		
rule	mean cov.	rel.	width	mean cov.	rel.	width	mean cov.	rel.	width
Theorem 3, Gaussian	0.952	1.000	0.3–0.4	0.953	1.000	0.3–0.5	0.951	1.000	0.3–0.4
Theorem 3, symm. unimodal	0.953	1.000	0.6–0.9	0.953	1.000	0.7–0.9	0.951	1.000	0.7–0.9
no shape, median zero	0.978	1.000	19.5–27.6	0.980	1.000	22.2–30.4	0.977	1.000	20.3–28.6
LatentCP	0.971	1.000	13.5–18.9	0.973	1.000	14.9–20.6	0.970	1.000	13.8–19.4
LatentCP, tuned	0.973	0.995	11.3–23.8	0.974	1.000	16.7–24.5	0.972	1.000	15.4–25.2
LatentCP, two levels	0.971	1.000	12.5–21.5	0.975	1.000	15.9–25.5	0.971	1.000	13.8–24.2
LatentCP, PAC rank	0.978	1.000	19.5–27.6	0.980	1.000	22.2–30.4	0.977	1.000	20.3–28.6
Markov rule	0.975	1.000	20.4–24.8	0.977	1.000	20.8–26.4	0.974	1.000	20.3–25.0
Fay–Herriot	0.915	0.960	-19.9 to -17.4	0.914	0.925	-20.1 to -16.7	0.914	0.905	-19.3 to -16.4
known lower bound	0.952	1.000	0.3–0.4	0.953	1.000	0.3–0.5	0.951	1.000	0.3–0.4
deconvolution, bin 0.01	0.870	0.030	-35.7 to -28.4	0.855	0.000	-42.8 to -30.4	0.860	0.020	-38.2 to -29.0
deconvolution, bin 0.2	0.907	0.600	-27.7 to -20.4	0.882	0.290	-32.3 to -18.9	0.900	0.550	-30.6 to -21.4

Tables S7–S9 give the full results behind Table 6. With 300 data sets, the binomial standard error of a reliability near 0.95 is 0.013, and with

Table S9: The t_3 control, outside $\mathcal{B}$: smallest mean latent coverage and smallest reliability over the three noise scenarios

	$K = 110$		$K = 1000$	
rule	mean cov.	rel.	mean cov.	rel.
Theorem 3, Gaussian	0.972	1.000	0.958	1.000
Theorem 3, symm. unimodal	0.972	1.000	0.959	1.000
no shape, median zero	0.989	1.000	0.979	1.000
LatentCP	0.976	1.000	0.974	1.000
LatentCP, tuned	0.975	0.993	0.974	1.000
LatentCP, two levels	0.977	0.997	0.974	1.000
LatentCP, PAC rank	0.989	1.000	0.979	1.000
Markov rule	0.984	1.000	0.975	1.000
Fay–Herriot	0.945	0.967	0.937	1.000
HetLDC	0.961	0.990	–	–
known lower bound	0.971	1.000	0.958	1.000
deconvolution, bin 0.01	0.437	0.000	0.875	0.150
deconvolution, bin 0.2	0.872	0.493	0.893	0.465

200 it is 0.015. The Fay–Herriot interval met its high-probability target for normal latent laws with Gaussian noise and fell short for the Laplace latent law and, at $K = 1000$, for normal latent laws with non-Gaussian noise. The PAC version of LatentCP coincides with the rule without a shape assumption to the precision shown: with $\gamma = \alpha/2$ its inversion keeps the w with $\mathrm{pr}(|w + e| \le t_\gamma) \ge 1/2$, which for symmetric noise is $|w|$ just below t_γ, and its high-probability rank at level $1 - \gamma = (1 + q)/2$ is that of the shape-free rule. With bin width 0.01 the deconvolution threshold fails at $K = 110$ because most bins are empty: its mean coverage is 0.28–0.43.

S6.5 Latent laws that differ between units

Calibration unit i has the latent law $W_i = b_i - X_1/u_i$, an exponential tail with rate u_i and $Q_q(|W_i|) = r_i = 1 + \tau_i/u_i$, so that τ_i measures the distance from the new unit's quantile $r_0 = 1$ in units of the tail scale, and Gaussian noise $\sigma_i Z$ with σ_i chosen adversarially, to maximize $\pi_i = \mathrm{pr}(|V_i| < r_0)$, which has a closed form. The latent reliability of rank k is $\mathrm{pr}(\sum_i X_i \le k-1)$ with independent $X_i \sim \mathrm{Bernoulli}(\pi_i)$. When the unit parameters are drawn independently from a design, the X_i are independent and identically distributed with mean $H = E\pi_i$, exact for the homogeneous and point-mass designs and a Monte Carlo mean over 4000 draws, with standard error about 2×10^{-4}, for the heterogeneous ones. The designs are: homogeneous extremal, $\tau_i = 0$ and $u_i = 200$; all units slightly easier than the new one, $r_i = 0.999995$, that is $\tau_i = -10^{-3}$ with $u_i = 200$ or $\tau_i = -10^{-2}$ with $u_i = 2000$; heterogeneous with the condition of Corollary 1, $u_i = 200L_i$ with L_i lognormal with log-scale 0.7 and $\tau_i = |N(0, 0.02^2)|$; the same with 10% of units easier, $\tau_i = -0.02$; and the new unit a random unit of the two-type population of point masses of § S3.2, with $\sigma = 10^{-4}$.

Table S10 shows three things. The bound of Corollary 1 is attained by homogeneous extremal laws, and heterogeneity among units that satisfy the condition only adds slack. The condition has no slack at the extremal

Table S10: Latent laws that differ between units, Gaussian noise: excess of the mean noisy coverage H over $\Psi_{\mathcal{N}}(q)$, and exact latent reliability of the usual rank, the rank of Theorem 3 and the rank without a shape assumption

			$K = 10^4$			$K = 10^6$		
q	design	$10^3\{H - \Psi_{\mathcal{N}}(q)\}$	usual	Theorem 3	no shape	usual	Theorem 3	no shape
0.8	homogeneous extremal ($\tau = 0$, $u = 200$)	0.00	0.713	0.951	1.000	0.000	0.950	1.000
	all units easier, $\tau = -10^{-3}$, $u = 200$	+0.76	0.645	0.928	1.000	0.000	0.398	1.000
	all units easier, $\tau = -10^{-2}$, $u = 2000$	+7.60	0.085	0.392	1.000	0.000	0.000	1.000
	heterogeneous, condition holds	-11.99	1.000	1.000	1.000	1.000	1.000	1.000
	10% of units easier, $\tau = -0.02$	-9.23	0.998	1.000	1.000	1.000	1.000	1.000
	new unit a random unit (mixture)	+95.49	0.000	0.000	0.955	0.000	0.000	0.976
0.9	homogeneous extremal ($\tau = 0$, $u = 200$)	0.00	0.918	0.952	1.000	0.150	0.950	1.000
	all units easier, $\tau = -10^{-3}$, $u = 200$	+0.88	0.864	0.914	1.000	0.000	0.094	1.000
	all units easier, $\tau = -10^{-2}$, $u = 2000$	+8.88	0.050	0.085	1.000	0.000	0.000	1.000
	heterogeneous, condition holds	-13.96	1.000	1.000	1.000	1.000	1.000	1.000
	10% of units easier, $\tau = -0.02$	-10.34	1.000	1.000	1.000	1.000	1.000	1.000
	new unit a random unit (mixture)	+49.15	0.000	0.000	0.956	0.000	0.000	0.970

law: lowering every r_i by a factor $1 - 5 \times 10^{-6}$ lowers the reliability of the rank of Theorem 3 to 0.094 at $K = 10^6$ and $q = 0.9$; whether a violation is small is relative to the latent tail scale, and with a tail ten times steeper the same violation gives 0.085 at $K = 10^4$. And when the new unit is a random unit of a heterogeneous population, the bi-log-concave ranks fail while the shape-free rank keeps its guarantee.

With one draw of the parameters of K units held fixed, the units are independent but not identically distributed, and the reliability is a Poisson-binomial probability, computed exactly by recursion. When the condition holds, $\max_i \pi_i$ equals $\Psi_{\mathcal{N}}(q)$ to five digits, attained by the units with τ_i near 0, and the reliability of the rank of Theorem 3 is 0.9989 ($K = 10^3$) and 1.0000 ($K = 10^4$) at $q = 0.9$, and 0.9955 and 1.0000 at $q = 0.8$. With 10% easier units, $\max_i \pi_i$ exceeds $\Psi_{\mathcal{N}}(q)$, 0.9187 at $q = 0.9$, but the

reliability stays at least 0.991, as the general form of § S3.2 allows.

S6.6 Contaminated latent laws

Each unit has the latent law $(1-\xi)G + \xi H$, with G the extremal law of § S6.1 for the target $q_0 = q/(1-\xi)$, an exponential tail with rate u/s and end just beyond the threshold, H the point mass at $1 + s\eta'$, and Gaussian noise $N(0, s^2)$, with $s = 10^{-3}$ and $\eta' = 10^{-6}$. Then $\mathrm{pr}(|W| \le 1) = q$, so $r = 1$, and the noisy coverage $H_\xi = \mathrm{pr}(|V_i| < 1)$ has a closed form; it is within 2×10^{-8} below $p_\xi(q)$. The latent reliability of rank k is $\mathrm{pr}\{\mathrm{Bin}(K, H_\xi) \le k - 1\}$, exactly.

Table S11: Contaminated extremal law, Gaussian noise: level $p_\xi(q)$ and exact latent reliability of the usual rank (u), the rank of Theorem 3 (T) and the rank at level $p_\xi(q)$ (c)

			$K = 10^3$			$K = 10^4$			$K = 10^5$		
q	ξ	$p_\xi(q)$	u	T	c	u	T	c	u	T	c
0.8	0%	0.80441	0.915	0.955	0.955	0.713	0.951	0.951	0.032	0.950	0.950
	0.5%	0.80667	0.883	0.936	0.954	0.497	0.861	0.950	0.000	0.438	0.950
	1%	0.80893	0.844	0.910	0.953	0.280	0.695	0.952	0.000	0.024	0.951
	2%	0.81348	0.741	0.835	0.958	0.040	0.257	0.952	0.000	0.000	0.951
	5%	0.82731	0.311	0.436	0.956	0.000	0.000	0.951	0.000	0.000	0.951
0.9	0%	0.90080	0.943	0.954	0.954	0.918	0.952	0.952	0.790	0.951	0.951
	0.5%	0.90322	0.907	0.925	0.952	0.720	0.804	0.953	0.039	0.181	0.950
	1%	0.90563	0.858	0.881	0.960	0.406	0.514	0.951	0.000	0.000	0.950
	2%	0.91049	0.707	0.745	0.955	0.027	0.049	0.953	0.000	0.000	0.951
	5%	0.92517	0.124	0.149	0.961	0.000	0.000	0.953	0.000	0.000	0.951

At $\xi = 0$ the rank of Theorem 3 and the contamination rank coincide. With $\xi > 0$, the rank of Theorem 3 behaves like the usual rank at $\xi = 0$: it is close to valid at $K = 10^3$ for $\xi \le 1\%$ and fails as K grows, while the

rank at level $p_\xi(q)$ keeps reliability at least 0.950 throughout and tends to it.

S6.7 Centred residuals: levels and ranks

Table S12 gives the excess $p^\zeta_{\mathcal{C}}(q) - q$ of Proposition 4, the calibration sizes from which the usual rank's exact latent reliability at the worst law falls below 0.9 and 0.5, and the ranks at $K = 10^4$ and 10^6 of the usual rule, the rule at level $p^\zeta_{\mathcal{C}}(q)$ and Theorem 3. At $K = 110$ the three ranks coincide in every case.

S6.8 Misspecified noise class

For a single standardized noise law ϵ and its scales, the one-end level is $\Psi_\epsilon(q) = \sup_{s>0} E\{\min(1, q\mathrm{e}^{-s\epsilon})\}$, attained along exponential tails by Proposition 2 applied to the class $\{s\epsilon : s > 0\}$; the two ends see ϵ and $-\epsilon$, so $H = \max(\Psi_\epsilon, \Psi_{-\epsilon})$ is a lower bound on any noisy level valid for $\mathcal{B}$ under this noise, and at the corresponding extremal latent law the count N of the proof of Theorem 3 is $\mathrm{Bin}(K, H)$ in the limit. Table S13 gives H at $q = 0.9$ and the exact latent reliability at that law of the ranks computed for the Gaussian, symmetric unimodal and mean-zero log-concave classes. Laws are standardized to mean zero and variance one; the two-point laws take

Table S12: Centred log-concave residuals, $\delta = 0.05$: excess of the level, sizes K at which the usual rank's reliability falls below 0.9 and 0.5, and ranks of the usual rule (u), the rule at level $p^{\zeta}_{\mathcal{C}}(q)$ (c) and Theorem 3 (T)

					$K = 10^4$			$K = 10^6$		
noise, q	ζ	excess	$K_{0.9}$	$K_{0.5}$	u	c	T	u	c	T
Gaussian, 0.8	0	1.7×10^{-4}	7.3×10^{5}	1.5×10^{7}	8067	8068	8110	800659	800829	805064
	0.05	2.7×10^{-4}	3.0×10^{5}	6.1×10^{6}	8067	8069	8110	800659	800925	805064
	0.1	3.9×10^{-4}	1.4×10^{5}	2.8×10^{6}	8067	8070	8110	800659	801050	805064
	0.2	7.4×10^{-4}	4.0×10^{4}	7.9×10^{5}	8067	8074	8110	800659	801398	805064
	0.3	1.2×10^{-3}	1.5×10^{4}	2.8×10^{5}	8067	8079	8110	800659	801895	805064
	0.5	3.3×10^{-3}	2.1×10^{3}	4.1×10^{4}	8067	8099	8110	800659	803933	805064
	≥ 1	4.4×10^{-3}	1.4×10^{3}	2.2×10^{4}	8067	8110	8110	800659	805064	805064
Gaussian, 0.9	0	2.8×10^{-6}	1.5×10^{9}	3.1×10^{10}	9050	9050	9058	900494	900497	901296
	0.05	1.1×10^{-5}	9.4×10^{7}	1.9×10^{9}	9050	9050	9058	900494	900505	901296
	0.1	3.1×10^{-5}	1.2×10^{7}	2.5×10^{8}	9050	9050	9058	900494	900525	901296
	0.2	1.3×10^{-4}	7.4×10^{5}	1.5×10^{7}	9050	9051	9058	900494	900621	901296
	0.3	4.1×10^{-4}	7.2×10^{4}	1.4×10^{6}	9050	9054	9058	900494	900906	901296
	0.5	7.4×10^{-4}	2.3×10^{4}	4.4×10^{5}	9050	9057	9058	900494	901233	901296
	≥ 1	8.0×10^{-4}	1.9×10^{4}	3.8×10^{5}	9050	9058	9058	900494	901296	901296
Gaussian, 0.95	0	1.1×10^{-7}	5.3×10^{11}	–	9537	9537	9538	950359	950359	950518
	0.05	1.6×10^{-6}	2.5×10^{9}	5.0×10^{10}	9537	9537	9538	950359	950361	950518
	0.1	7.0×10^{-6}	1.3×10^{8}	2.7×10^{9}	9537	9537	9538	950359	950366	950518
	0.2	6.9×10^{-5}	1.3×10^{6}	2.7×10^{7}	9537	9537	9538	950359	950428	950518
	0.3	1.1×10^{-4}	4.9×10^{5}	9.8×10^{6}	9537	9538	9538	950359	950473	950518
	0.5	1.5×10^{-4}	2.8×10^{5}	5.6×10^{6}	9537	9538	9538	950359	950510	950518
	≥ 1	1.6×10^{-4}	2.6×10^{5}	5.1×10^{6}	9537	9538	9538	950359	950518	950518
symm. unimodal, 0.8	0	4.7×10^{-4}	1.0×10^{5}	2.0×10^{6}	8067	8071	8143	800659	801125	808374
	0.05	6.9×10^{-4}	4.6×10^{4}	9.0×10^{5}	8067	8073	8143	800659	801352	808374
	0.1	9.7×10^{-4}	2.3×10^{4}	4.6×10^{5}	8067	8076	8143	800659	801630	808374
	0.2	1.7×10^{-3}	7.8×10^{3}	1.5×10^{5}	8067	8083	8143	800659	802349	808374
	0.3	2.7×10^{-3}	3.2×10^{3}	6.0×10^{4}	8067	8093	8143	800659	803349	808374
	0.5	6.4×10^{-3}	6.4×10^{2}	1.1×10^{4}	8067	8130	8143	800659	807053	808374
	≥ 1	7.7×10^{-3}	4.2×10^{2}	7.3×10^{3}	8067	8143	8143	800659	808374	808374
symm. unimodal, 0.9	0	1.3×10^{-5}	6.7×10^{7}	1.4×10^{9}	9050	9050	9068	900494	900508	902278
	0.05	4.6×10^{-5}	5.5×10^{6}	1.1×10^{8}	9050	9051	9068	900494	900541	902278
	0.1	1.1×10^{-4}	9.0×10^{5}	1.8×10^{7}	9050	9051	9068	900494	900609	902278
	0.2	4.0×10^{-4}	7.9×10^{4}	1.5×10^{6}	9050	9054	9068	900494	900893	902278
	0.3	1.1×10^{-3}	1.0×10^{4}	1.9×10^{5}	9050	9061	9068	900494	901616	902278
	0.5	1.7×10^{-3}	4.4×10^{3}	8.6×10^{4}	9050	9067	9068	900494	902177	902278
	≥ 1	1.8×10^{-3}	4.0×10^{3}	7.6×10^{4}	9050	9068	9068	900494	902278	902278
symm. unimodal, 0.95	0	6.9×10^{-7}	1.3×10^{10}	2.7×10^{11}	9537	9537	9541	950359	950360	950789
	0.05	8.1×10^{-6}	9.5×10^{7}	1.9×10^{9}	9537	9537	9541	950359	950367	950789
	0.1	3.1×10^{-5}	6.5×10^{6}	1.3×10^{8}	9537	9537	9541	950359	950390	950789
	0.2	2.3×10^{-4}	1.2×10^{5}	2.4×10^{6}	9537	9539	9541	950359	950589	950789
	0.3	3.4×10^{-4}	6.0×10^{4}	1.1×10^{6}	9537	9540	9541	950359	950695	950789
	0.5	4.1×10^{-4}	3.9×10^{4}	7.5×10^{5}	9537	9541	9541	950359	950772	950789
	≥ 1	4.3×10^{-4}	3.6×10^{4}	6.9×10^{5}	9537	9541	9541	950359	950789	950789

Rows with $\zeta \geq 1$ are $\Psi_{\mathcal{C}}(q) - q$ (proved); the others are numerical (§ S3.4). $K_{0.9}$ and $K_{0.5}$ from an exact scan of the binomial reliability; – where beyond the scan.

the values ± 1, and -0.5 and 2 with probabilities 0.8 and 0.2. A value of H below a class level does not prove that the class rank is valid under that noise, because H is only a lower bound for laws outside the class.

Table S13: Misspecified noise class at $q = 0.9$, $\delta = 0.05$: lower bound H on the needed level for single noise laws, exact latent reliability at $K = 10^4$ at the extremal law of the ranks for the Gaussian (G), symmetric unimodal (SU) and mean-zero log-concave (LC, level 0.906) classes, and the size K from which the Gaussian rank's reliability is below 0.5

noise law	class	H	G	SU	LC	K, G below 0.5
Gaussian	Gaussian	0.90080	0.952	0.977	1.000	–
uniform	symmetric unimodal	0.90179	0.909	0.953	0.999	250,137
Laplace	symmetric unimodal	0.90038	0.964	0.984	1.000	–
logistic	symmetric unimodal	0.90055	0.960	0.982	1.000	–
t5	symmetric unimodal	0.90038	0.964	0.984	1.000	–
centred exponential	mean-zero log-concave	0.90518	0.576	0.703	0.974	12,693
centred gamma(2)	mean-zero log-concave	0.90317	0.809	0.888	0.996	43,328
centred gamma(5)	mean-zero log-concave	0.90194	0.900	0.948	0.999	187,748
skew-normal(4)	mean-zero log-concave	0.90186	0.905	0.951	0.999	218,406
centred lognormal(0.5)	mean zero	0.90303	0.820	0.896	0.996	49,097
centred lognormal(1)	mean zero	0.90932	0.108	0.185	0.702	3,365
centred chi-square(1)	mean zero	0.90906	0.125	0.210	0.732	3,628
two-point symmetric	median zero	0.90500	0.599	0.724	0.978	13,817
two-point, p = 0.2	mean zero	0.91810	0.000	0.000	0.004	776

Within the symmetric unimodal class, the Laplace, logistic and t_5 laws stay below the Gaussian level, and only the uniform law, the extremal one, exceeds it. Skewed log-concave laws exceed both symmetric levels and are covered by the log-concave level. Strongly skewed laws outside the log-concave class, such as a centred lognormal law with log-scale 1, a centred chi-square law with one degree of freedom and a skewed two-point law, exceed even that level; such laws approximate the sampling error of a mean of a few units from a skewed finite population. These are extremal latent

laws; for centred residuals the excess is much smaller, as in § S6.7.

S6.9 Between the extremal and the benign designs

With units independent and identically distributed, $r = Q_q(|W|)$ and $H = \mathrm{pr}(|V| < r)$, the count N of the proof of Theorem 3 is $\mathrm{Bin}(K, H)$, so the latent reliability of rank k is $\mathrm{pr}\{\mathrm{Bin}(K, H) \le k - 1\}$ exactly; if $H \le q$ the usual rank is valid at every K. H is computed by quadrature as $q + \int f(w)\{\kappa(w) - 1(|w| < r)\}\, dw$, with $\kappa(w) = \mathrm{pr}(|w+e| < r)$ the noise kernel, which avoids cancellation in $H - q$; values agree with 30-digit quadrature to 3×10^{-17} and, for the proportions, with Monte Carlo.

Designs. Latent laws, standardized and all log-concave: symmetric (normal, Laplace); smooth and skewed (centred gamma with shapes 2 and 5, skew-normal with shape 4, and their reflections); J-shaped, with the mode at an end of the support (centred exponential and half-normal, and their reflections). A bias b is added so that $\zeta = |E(W)|/Q_q(|W|)$ is 0, 0.1, 0.2, 0.3, 0.5 or 0.75. Noise is Gaussian or uniform with variance ratio $\rho = \sigma^2/\mathrm{var}(W)$ in $\{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 0.58, 1, 3\}$, equal for all units or heterogeneous, $\sigma^2 = \rho L/E(L)$ with L lognormal with log-scale 0.7. Proportions: $\theta \sim \mathrm{Beta}(a, b)$ for six log-concave choices with means 0.1 to 0.5 and for Beta(0.5, 4.5), which is not bi-log-concave; $W = \theta - c$ with c the

mean or the mean ± half a standard deviation; noise $N\{0, \theta(1-\theta)/n\}$ with $n \in \{20, 50, 200, 1000\}$, which depends on the latent value and so violates the independence assumed by Theorem 3. Rules: the usual rank and the rank of Theorem 3 for the noise class; $q = 0.8$, 0.9, 0.95; reliabilities for every K from 29 to 3000 and on a fine grid up to 10^6.

Table S14: Exact reliability between the extremal and the benign designs, $\delta = 0.05$: number of designs, number with $H > q$, largest $H - q$ ($\times 10^4$), and smallest latent reliability over $K \leq 10^5$ of the usual rank and of the rank of Theorem 3

latent laws	q	designs	$H > q$	$\max(H - q)$	usual	Theorem 3
symmetric	0.8	432	0	−1.2	0.951	0.956
	0.9	432	0	−1.0	0.951	0.952
	0.95	432	0	−0.5	0.950	0.952
skewed smooth	0.8	1296	0	−0.8	0.950	0.956
	0.9	1296	0	−0.6	0.950	0.952
	0.95	1296	0	−0.3	0.950	0.952
J-shaped	0.8	864	87	67.4	0.000	0.951
	0.9	864	30	13.5	0.588	0.950
	0.95	864	8	4.0	0.856	0.950
proportions	0.8	72	0	−6.3	0.952	0.957
	0.9	72	0	−5.8	0.952	0.953
	0.95	72	0	−4.9	0.952	0.953
proportions (not log-concave)	0.8	12	0	−2.7	0.951	0.956
	0.9	12	0	−3.1	0.951	0.953
	0.95	12	0	−2.6	0.951	0.952

Table S14 shows three things for the designs studied; they are findings, not theorems. For the symmetric and smooth skewed residuals, biased up to $\zeta = 0.75$, $H < q$ at every noise level: by the small-noise expansion of § S3.4, $H - q$ has the sign of $f'(r) - f'(-r)$, which is negative when the mode lies inside $(-r, r)$, and larger noise spreads the scores further. With noise

variance at least a tenth of the latent one and $\zeta \le 0.5$, $H < q$ for every latent law of the study, with $H - q \le -2.5 \times 10^{-3}$. The usual rank loses reliability only for J-shaped residuals, whose density rises to an edge just beyond the threshold, under small noise, $\rho \le 0.03$; there the rank of Theorem 3 keeps it. For proportions, the noise variance $\theta(1-\theta)/n$ vanishes at the edge of a J-shaped Beta law, which keeps $H < q$; with independent noise of the same mean variance the not log-concave Beta$(0.5, 4.5)$ law exceeds even $\Psi_{\mathcal{N}}(q)$, by 1.6×10^{-3} at $q = 0.8$ and $n = 1000$.

S7. The school-district application

Data. The Academic Performance Index population `apipop` is taken from the CRAN package `survey`, version 4.5, checked against a SHA-256 digest. It has one record per California school; districts with at least five schools, 325 of them, form the population. Covariates are the socio-demographic variables (shares of students on subsidised meals, English language learners and new to the school; parents' education in five categories and on average; shares of fully and emergency credentialed teachers; enrolment; response rate), a year-round indicator and the school type; the second set adds the 1999 index. Missing covariates are replaced by their medians.

Design. Each replication splits the districts at random into 110 train-

ing, 110 calibration and 105 evaluation districts and samples $n = 2$ or 3 schools per training and calibration district without replacement. A ridge regression of the 2000 index on the covariates, with the penalty chosen by cross-validation over 30 values, is fitted to the sampled schools of the training districts. For a calibration district c the score V_c is the mean residual of its sampled schools and $\hat{D}_c = (1 - n/N_c)s_c^2/n$, with s_c^2 the sample variance of those residuals and N_c the number of schools in the district, estimates its design variance with $n - 1$ degrees of freedom. The target of an evaluation district is the mean residual over all its schools, its true mean minus the population mean of the predictor, and its interval is centred at zero on this scale except for the Fay–Herriot interval, centred at the fitted mean. Coverage is the share of the 105 evaluation districts covered, a finite-population coverage. There are four conditions, the two covariate sets with $n = 2$ and 3, and 120 replications each.

Rules. The ranks of Theorem 3 for the three bi-log-concave rows of Table 2 are all 105 at $K = 110$, equal to the usual rank, and use no variance estimate; the rank without a shape assumption is 109. The Fay–Herriot interval, LatentCP and the Markov rule plug in the $\hat{D}_c$ as if known, a case that no guarantee covers. LatentCP needs the variance of the new district, which is unknown; in this application variant it is drawn from the

calibration estimates, 200 draws per replication.

Table S15: School districts: mean and smallest coverage of the evaluation districts over 120 replications, share of replications with coverage at least 0.9, mean width, and width relative to the usual rank (%)

rule	mean cov.	min. cov.	share ≥ 0.9	width	vs usual
with 1999 index, $n = 3$					
usual rank	0.980	0.914	1.000	72.3	0.0
Theorem 3 (all three classes)	0.980	0.914	1.000	72.3	0.0
no shape, median zero	0.995	0.971	1.000	93.2	28.8
Fay–Herriot, plug-in $\hat{D}_c$	0.944	0.848	0.908	58.6	-19.0
LatentCP, plug-in $\hat{D}_c$	0.984	0.933	1.000	76.0	5.0
Markov rule, plug-in $\hat{D}_c$	0.990	0.952	1.000	84.0	16.1
with 1999 index, $n = 2$					
usual rank	0.989	0.952	1.000	88.1	0.0
Theorem 3 (all three classes)	0.989	0.952	1.000	88.1	0.0
no shape, median zero	0.998	0.971	1.000	119.3	35.5
Fay–Herriot, plug-in $\hat{D}_c$	0.967	0.895	0.992	70.0	-20.6
LatentCP, plug-in $\hat{D}_c$	0.990	0.948	1.000	93.2	5.7
Markov rule, plug-in $\hat{D}_c$	0.993	0.962	1.000	98.9	12.2
socio-demographic only, $n = 3$					
usual rank	0.973	0.914	1.000	150.4	0.0
Theorem 3 (all three classes)	0.973	0.914	1.000	150.4	0.0
no shape, median zero	0.993	0.962	1.000	198.1	31.6
Fay–Herriot, plug-in $\hat{D}_c$	0.941	0.876	0.925	126.7	-15.8
LatentCP, plug-in $\hat{D}_c$	0.978	0.923	1.000	157.4	4.6
Markov rule, plug-in $\hat{D}_c$	0.988	0.943	1.000	175.1	16.4
socio-demographic only, $n = 2$					
usual rank	0.983	0.933	1.000	168.1	0.0
Theorem 3 (all three classes)	0.983	0.933	1.000	168.1	0.0
no shape, median zero	0.997	0.962	1.000	225.4	34.1
Fay–Herriot, plug-in $\hat{D}_c$	0.958	0.886	0.975	140.0	-16.7
LatentCP, plug-in $\hat{D}_c$	0.986	0.938	1.000	177.1	5.4
Markov rule, plug-in $\hat{D}_c$	0.992	0.952	1.000	193.6	15.2

Table S15 gives the results. The rank of Theorem 3 covered at least 90% of the evaluation districts in all 480 replications. The Fay–Herriot interval was shorter, but covered fewer than 90% of the districts in up to 9.2% of the replications of a condition; LatentCP was 4.6–5.7% wider than

the rank's interval and the Markov rule 12–16% wider. The guarantee of Theorem 3 still needs the noise of the direct estimates to lie in the stated class, independent of the latent residuals, and the calibration districts to be no easier than a new one in the sense of Corollary 1; neither can be confirmed from these data. With two or three schools, the direct estimates are means of very few residuals, so their noise need not be close to Gaussian, which is one reason to state the guarantee for the symmetric unimodal or log-concave class; at $K = 110$ the rank is the same.

S8. Reproducibility

All computations are in the public repository `https://github.com/kunhailP/noisy-calibration-latent-coverage`, with exact package versions in `requirements-lock.txt`. The package `latentcov` provides the ranks (`rank`, `rank_for_level`), the levels by noise class and latent shape (`levels.level`), the Gaussian closed form and its enclosure (`gaussian`), the certificates (`interval`, `noise_classes`), the simulation laws (`laws`) and the rules that use known variances (`kv`). The test suite, run by `make test`, checks the closed form of Theorem 2 against direct maximization, the ranks of Table 2, the extremal laws, the class memberships of the simulation laws and Lemmas S5 and S6 on random probability vectors.

Table S16 maps results to scripts and outputs; the tables of this Supplement are written from those outputs by `experiments/supp_tables.py`. The Makefile sets one BLAS thread per process.

Table S16: Results, scripts in `experiments` and outputs in `results`

result	script	output
Figure 1, Table S4	`r01_reliability_curves.py`	`reliability_curves.csv`, `reliability_summary.json`
Table 5, Table S6	`r02_same_information.py`	`same_information_v2*.csv`
mixed noise, Table S5	`r03_mixed_noise_stress.py`	`mixed_noise_stress*.csv`
Table 6, Tables S7–S9	`r04_competitors.py`	`competitors_v2_hetldc*.csv`
application, Table S15	`fetch_apipop.py`, `r05_apipop.py`	`apipop*.csv`
Corollary 1, Table S10	`r06_heterogeneous_latent.py`	`heterogeneous_latent*.csv`
Proposition 5, Table S11	`r07_contamination.py`, `latentcov.contamination`	`contamination*.csv`
Proposition 4, Tables 4 and S12	`r08_centred.py`, `latentcov.centred`	`centred_*.csv`
misspecified noise class, Table S13	`r09_noise_misspec.py`, `latentcov.misspec`	`noise_misspec.csv`
intermediate designs, Table S14	`r10_intermediate.py`, `latentcov.intermediate`	`intermediate*.csv`
levels and ranks of Table 2	`latentcov.rank`	computed on call
certificates for the noise classes and c_q	`latentcov.noise_classes`, `latentcov.interval`	`legacy/*.json`

References

Anderson, T. W. (1955). The integral of a symmetric unimodal function over a symmetric convex set and some probability inequalities. *Proc. Amer. Math. Soc. 6*, 170–176.

Bobkov, S. G. and G. P. Chistyakov (2015). On concentration functions of random variables. *J. Theoret. Probab. 28*, 976–988.

Chatterjee, S., P. Lahiri, and H. Li (2008). Parametric bootstrap approximation to the distri-

bution of EBLUP and related prediction intervals in linear mixed models. *Ann. Statist. 36*, 1221–1245.

Cohen, Y., J. Goldberger, and T. Tirer (2025). Efficient conformal prediction for regression models under label noise. *arXiv preprint arXiv:2509.15120*.

Darroch, J. N. (1964). On the distribution of the number of successes in independent trials. *Ann. Math. Statist. 35*, 1317–1321.

Dharmadhikari, S. and K. Joag-dev (1988). *Unimodality, Convexity, and Applications*. Boston: Academic Press.

Dümbgen, L., P. Kolesnyk, and R. A. Wilke (2017). Bi-log-concave distribution functions. *J. Statist. Plann. Inference 184*, 1–17.

Fradelizi, M. and O. Guédon (2004). The extreme points of subsets of s-concave probabilities and a geometric localization theorem. *Discrete Comput. Geom. 31*, 327–335.

Fradelizi, M. and O. Guédon (2006). A generalized localization theorem and geometric inequalities for convex bodies. *Adv. Math. 204*, 509–529.

Guille-Escuret, C. and E. Ndiaye (2024). From conformal predictions to confidence regions. *arXiv preprint arXiv:2405.18601*.

Hoeffding, W. (1956). On the distribution of the number of successes in independent trials. *Ann. Math. Statist. 27*, 713–721.

Jochmans, K. and M. Weidner (2024). Inference on a distribution from noisy draws. *Econometric*

Theory 40, 60–97.

Johansson, F. (2017). Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Trans. Comput. 66*, 1281–1292.

Kannan, R., L. Lovász, and M. Simonovits (1995). Isoperimetric problems for convex bodies and a localization lemma. *Discrete Comput. Geom. 13*, 541–559.

Prékopa, A. (1973). On logarithmic concave measures and functions. *Acta Sci. Math. (Szeged) 34*, 335–343.

Sampford, M. R. (1953). Some inequalities on Mill's ratio and related functions. *Ann. Math. Statist. 24*, 130–132.

Saumard, A. (2019). Bi-log-concavity: some properties and some remarks towards a multi-dimensional extension. *Electron. Commun. Probab. 24*(61), 1–8.

Steinbrecher, G. and W. T. Shaw (2008). Quantile mechanics. *European J. Appl. Math. 19*, 87–112.

Zheng, M., W. Zhou, and S. Zhu (2026). Beyond prediction: conformal inference for latent distributional parameters. *arXiv preprint arXiv:2608.03607*.